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REVIEW ARTICLE

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Adaptive social manufacturing: a human-centric, resilient, and sustainable framework for advancing Industry 5.0

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ABSTRACT

This paper presents the Adaptive Social Manufacturing (ASM) as a strategic framework for integrating emerging technologies, such as artificial intelligence, into manufacturing in a socially responsible, economically viable, and practically feasible manner. Building on Industry 4.0 foundations, ASM addresses the need for a balanced, human-centric framework that aligns technological advancements with the ethical, social, and environmental values central to Industry 5.0. The framework introduces core enabling capabilities, such as Digitalization Strategic Management and Resource Readiness, to drive responsible technology adoption, while preventive capabilities, including Technology Governance and Sustainability Performance Management Systems, mitigate potential socio-environmental risks. To validate the feasibility and practicality of the framework, the study incorporates a longitudinal case study of a medium-sized healthcare product manufacturer. This case demonstrates how ASM facilitated real-time monitoring, process integration, and sustainable innovation, achieving measurable operational improvements while balancing economic, social, and environmental objectives. The findings highlight how ASM enables feasible, stepwise digital transformation, ensuring that economic gains are leveraged to promote social inclusion and environmental responsibility. ASM offers a structured pathway for manufacturers seeking to transition beyond efficiency-driven digitalisation toward value-driven innovation, and holds promise as a guiding framework for policy design, workforce development, and cross-sectoral collaboration in future industrial ecosystems.

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Industry 5.0; Industry 4.0; digitalisation; human-centric; sustainability; resilience

1. Introduction

Industry 5.0 and the shift toward human-centricity is a recent and significant development that has gained substantial attention (Maddikunta et al. 2022). This is notable given that Industry 4.0 is still evolving, with manufacturers investing in emerging technologies such as Artificial Intelligence (AI) and the Industrial Internet of Things (IIoT) (Yang et al. 2023). The shift to Industry 5.0 reflects concerns that Industry 4.0 has often failed to address, or even worsened, social and environmental issues (Rana et al. 2022). While Industry 4.0 has brought many economic gains (Sarbu 2022), it has also negatively impacted employment, social inequality, and individual autonomy (Agarwal et al. 2022; Passalacqua et al. 2025). Socially, the technology-driven nature of Industry 4.0 transformation has led to job polarisation, skill gaps, and technostress, disproportionately affecting lower-skilled workers (Grybauskas, Stefanini, and Ghobakhloo 2022). Moreover, the absence of comprehensive governance has resulted in fragmented sustainability efforts, with economic gains often prioritised over environmental stewardship and workforce well-being (Ghobakhloo et al. 2024). Under such circumstances, Industry 5.0 arises as a response to the shortcomings of Industry 4.0 in fixating on productivity and economic efficiency, often at the expense of broader social issues (Grosse et al. 2023). In the same vein, the European Commission's 2022 policy highlights that Industry 4.0 is no longer sufficient to promote a sustainable future for European industry (Renda et al. 2022).

At the core of Industry 4.0 lies smart manufacturing. Smart manufacturing emphasises the integration of advanced technologies to enhance production efficiency and optimise processes (Singh, Madaan, and Kumar 2023) while also reshaping traditional business models and operations (Wen et al. 2023). However, despite its contributions, smart manufacturing is associated with two critical shortcomings in the Industry 4.0 framework.

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First, smart manufacturing's emphasis on productivity often overlooks societal values, especially in manufacturers without strong corporate social responsibility (Ghobakhloo et al. 2023a). As a result, the push for digitalisation has shaped workforce dynamics, sometimes to the detriment of employees' career development and well-being (Agarwal et al. 2022). This profit-driven orientation created a disconnect between manufacturing systems and societal and environmental well-being, which are pressing concerns in the modern world (Grybauskas, Stefanini, and Ghobakhloo 2022). Furthermore, the rigid focus on technological advancements without equal emphasis on human and ecological considerations has hindered the ability of manufacturing systems to adapt to the rapidly changing socio-economic landscape (Chiarini 2021). As global challenges like regional conflicts, climate change, and rising social inequalities become urgent (Agrawal et al. 2024), the smart manufacturing model of Industry 4.0 has proven inadequate to address these more significant systemic issues (Xu et al. 2021).

These two shortcomings highlight why Industry 5.0 critiques Industry 4.0's narrow focus on technology and productivity (Abdel-Basset, Mohamed, and Chang 2025). As a result, scholars and socio-industrial experts argue that the smart manufacturing model must evolve to prioritise human needs over technological advancement, in line with Industry 5.0's human-centric philosophy (Zhang et al. 2023). Additionally, it should emphasise adaptability and resilience, enabling manufacturing systems to handle disruptions while advancing environmental sustainability (Sindhwani et al. 2024). This new framework termed Adaptive Social Manufacturing (ASM) in this study, is an evolution of the smart manufacturing model that aligns with Industry 5.0's goals by incorporating social values and adaptive resilience into its operational framework (Renda et al. 2022).

Despite its relevance, the concept of ASM remains theoretically underdeveloped and empirically unvalidated, primarily because Industry 5.0 itself is still in its formative stages (Panagou, Neumann, and Fruggiero 2024). This study addresses this gap by taking a foundational step toward defining, developing, and validating the ASM concept. It explores how ASM can be delineated as a practical manufacturing framework that leverages advanced digital technologies, governance mechanisms, and human-centric approaches to create sustainable, adaptive manufacturing ecosystems. To operationalise the framework and demonstrate its feasibility, the study incorporates a longitudinal case study of a medium-sized healthcare product manufacturer, which implemented ASM principles across multiple transformation phases. The case highlights how ASM facilitated real-time monitoring, process integration, and sustainable innovation, ensuring that economic gains were leveraged to advance social and environmental objectives.

By developing and validating the ASM framework, this study provides valuable contributions to both research and practice. The conceptual insights can stimulate interdisciplinary collaborations, integrating evolving societal dynamics into manufacturing processes under Industry 5.0. Meanwhile, the empirical findings offer pragmatic guidance for manufacturers seeking to implement sustainable digital transformation strategies, fostering resilient, human-centred operational models that align technological progress with societal well-being.

2. Methodology

The study used a dual-objective, content-centric review to explore the literature and develop the ASM framework. This approach provided both a broad understanding of how Industry 4.0 has influenced sustainability in manufacturing and a focused identification of the key concepts, technologies, and principles underlying ASM. The methodological process followed the steps shown in Figure 1 and was organised around two systematic literature review (SLR) objectives. The first objective examined the sustainability impact of Industry 4.0 in manufacturing. The second extracted ASM-related concepts, including its technologies, principles, and objectives.

The first objective aimed to explain the emergence of ASM by assessing how Industry 4.0 has performed in terms of sustainability. While Industry 4.0 has improved manufacturing efficiency, its impact on sustainability has been inconsistent. Reported benefits are often accompanied by negative consequences, especially in relation to social inclusion and environmental outcomes (e.g. Dieste et al. 2024; Ghobakhloo et al. 2024). To examine this, a systematic search was conducted in Scopus and Web of Science using a predefined string with terms related to Industry 4.0, sustainability, environment, and social or human factors. This search returned 1,281 documents. These were screened based on criteria that excluded non-English texts, documents with no full-text access, and studies lacking practical insights into Industry 4.0 and sustainability. This narrowed the list to 391 documents.

To expand coverage, backward and forward review strategies were applied. The backward review examined references within the 391 documents and identified 103 additional candidates. The forward review used Google Scholar to find 130 newer documents citing the shortlisted studies. All new documents were screened again using the same criteria, resulting in a final set of 461 documents. These formed the basis for in-depth content analysis, where documents were coded and categorised

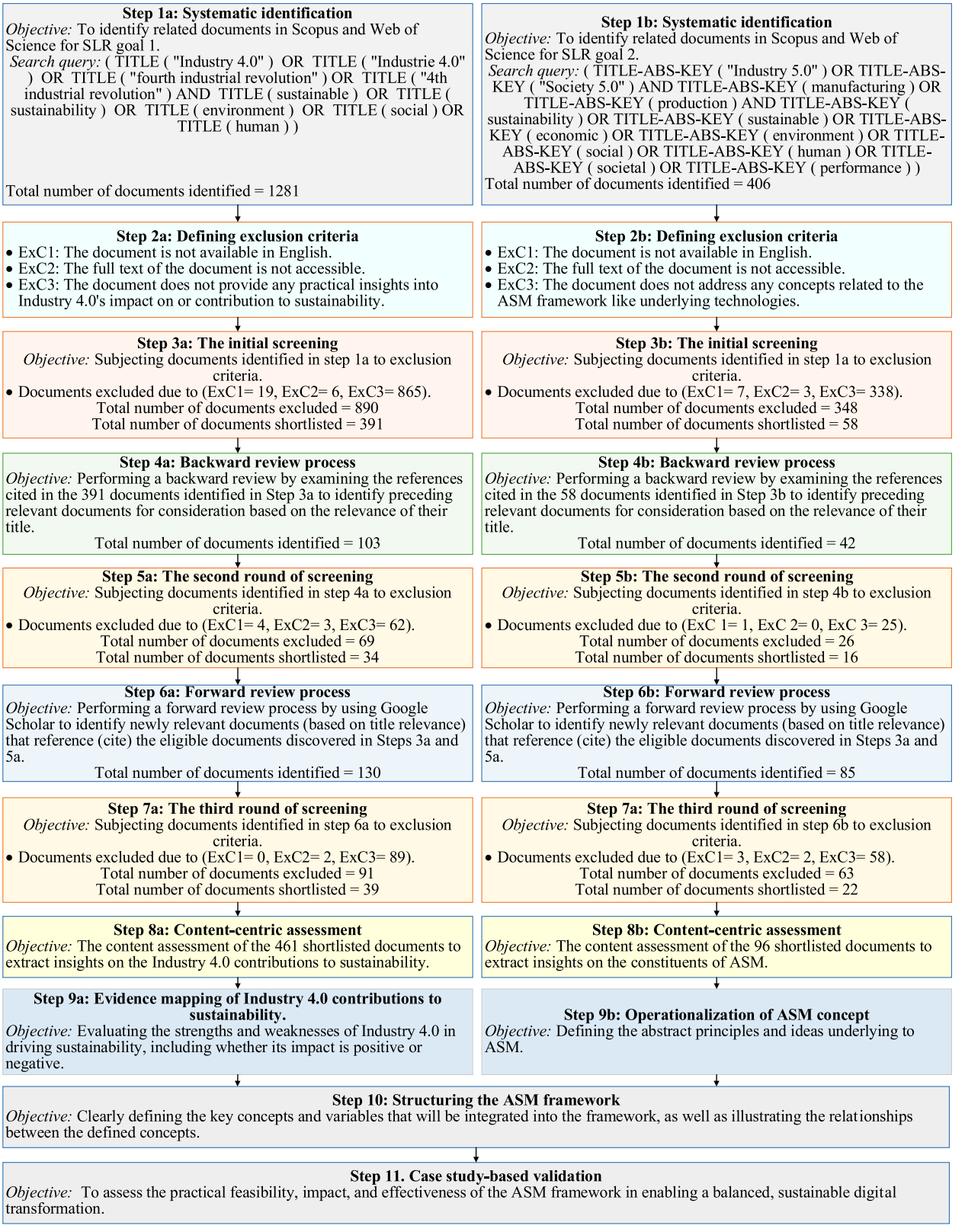


Figure 1. Steps undertaken for the content-centric review of the literature (Source: Authors).

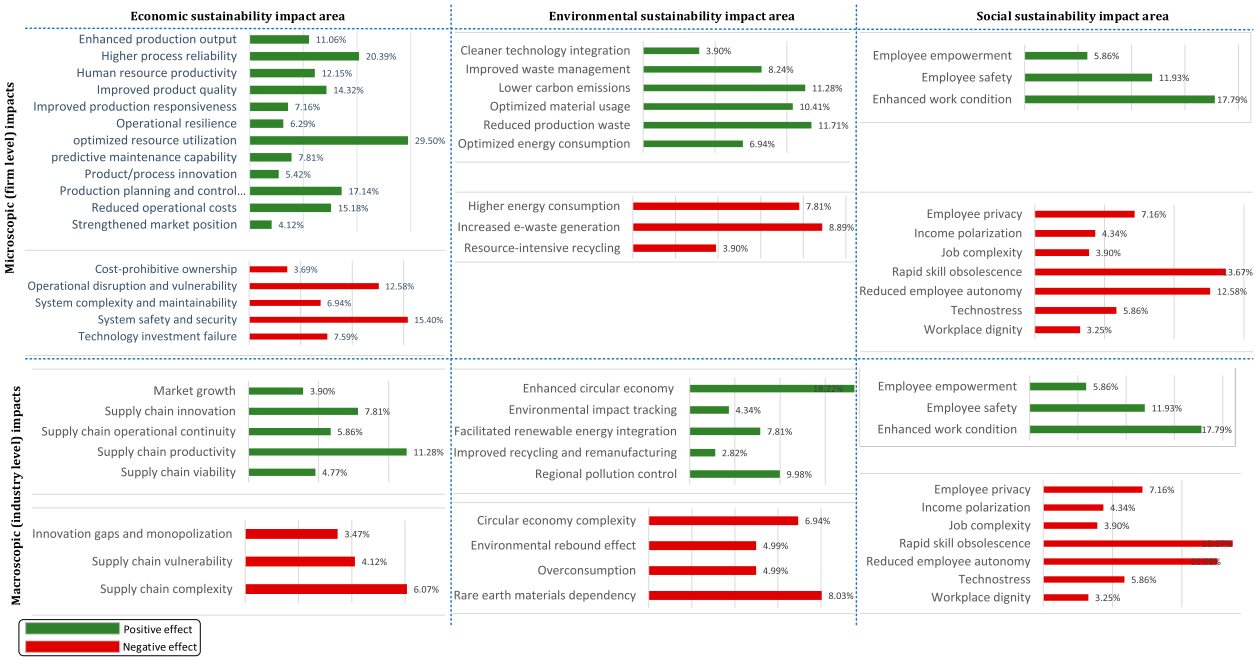


Figure 2. Sustainability impact of Industry 4.0 in the manufacturing context (Source: Authors).

based on whether the sustainability impacts of Industry 4.0 were mostly positive, negative, or mixed across environmental, social, and economic dimensions.

The second objective focused on identifying core concepts, technologies, and principles needed to build the ASM framework. This was shaped by the findings from the first review, which showed gaps in Industry 4.0's sustainability performance and the need for a more adaptive, socially conscious model. A new systematic search was conducted in Scopus and Web of Science using terms related to Industry 5.0, manufacturing, sustainability, and performance. This search produced 406 documents. After applying the same exclusion criteria, the list was reduced to 58 relevant documents. Backward and forward reviews added 42 and 85 more documents, respectively. Following a second screening, 96 documents were retained for detailed content analysis. These were coded to identify enabling technologies, social and sustainability principles, and ASM's strategic goals.

To ensure reliability, the content analysis (steps 8a and 8b) followed established guidelines (Drisko and Maschi 2016). Two independent assessors applied a standardised assessment protocol to minimise bias. Disagreements were resolved through discussion and joint review, ensuring consistent categorisation. This process produced key insights for both review objectives. For the first, it mapped Industry 4.0's sustainability performance, identifying both contributions and limitations. For the second, it established the essential components, technologies, and principles that form the foundation of the ASM framework.

The study further moved to the final phase, where results from both SLR objectives were combined to construct the ASM framework. This included defining ASM's core components, mapping their relationships, and showing how they respond to Industry 4.0 challenges while aligning with Industry 5.0 goals. A longitudinal case study was then conducted to validate the framework. It documented ASM's application across several transformation stages at a case company, assessing its feasibility, effectiveness, and sustainability impact.

3. From industry 4.0 to ASM

ASM has emerged as a response to the limitations of Industry 4.0's smart manufacturing model, especially its inconsistent sustainability outcomes. While Industry 4.0 has transformed manufacturing through technologies like AI, IoT, and cyber-physical systems, its impact on sustainability has been uneven (Dieste et al. 2024). As shown in Figure 2 and confirmed by the systematic review under SLR objective one, both positive and negative sustainability effects have been identified across economic, environmental, and social dimensions. Industry 4.0 offers the potential to enhance sustainability at both firm and sector levels. It enables more efficient production, leaner supply chains, and cleaner operations (Narkhede et al. 2024a). Advances in predictive maintenance, process optimisation, and energy-efficient production have improved factory-level sustainability (Samadhiya, Agrawal, and Garza-Reyes 2024). Reduced waste, optimised resource use, and cleaner technologies

have contributed to environmental gains at the micro level (Rossini et al. 2022). At the macro level, Industry 4.0 has supported circular economy practices and integration of renewables into manufacturing (Mastrocinque et al. 2022).

However, these benefits are often undermined by negative externalities. High energy use from IoT devices and real-time systems has become a growing concern (Kunkel et al. 2023). E-waste from rapidly obsolete hardware adds to the environmental burden (Shittu, Williams, and Shaw 2021), while reliance on rare earth materials and resource-intensive recycling strains global supply chains (Meinhold, Wagner, and Dhar 2024). These issues often stem from weak governance and strategic mismanagement. Without effective sustainability oversight, the gains from Industry 4.0 remain fragile and incomplete (Narkhede et al. 2025).

The most serious limitations appear in the social dimension, which reinforces the need for ASM. As shown in Figure 2, smart manufacturing under Industry 4.0 has prioritised productivity over workforce well-being and equity (Nara et al. 2021). While safety has improved and some employees have benefited from upskilling (Zorzenon, Lizarelli, and de A. Moura 2022), these gains are overshadowed by widespread skill obsolescence, job complexity, technostress, and declining autonomy (Dieste et al. 2024; Ghobakhloo et al. 2024; Grybauskas, Stefanini, and Ghobakhloo 2022). Traditional roles have been displaced or reshaped into precarious, tech-dependent positions (Hoque et al. 2022), while income gaps and unequal job access have widened social divides (Silva et al. 2022). At the broader societal level, automation has displaced many low – and middle-skilled jobs, while high-tech employment has favoured those with advanced education (Grybauskas, Stefanini, and Ghobakhloo 2022; Hoque et al. 2022). This deepens digital inequality and stratifies opportunity along socio-economic lines (Castelo-Branco et al. 2023). Unchecked smart manufacturing can further erode data privacy, social inclusion, and wealth equality (Saura, Ribeiro-Soriano, and Palacios-Marqués 2022).

These combined social and environmental challenges demonstrate the need for ASM as an evolution of smart manufacturing. ASM shifts the focus toward a more balanced, inclusive, and human-centred approach that aligns technological progress with broader goals of sustainability and equity.

4. ASM operational definition

ASM is a digital transformation framework that integrates technologies such as AI, IIoT, and cyber-physical systems within a human-centric and sustainability-focused manufacturing model. Recognising that these technologies have both positive and negative effects, ASM is built on responsible technology governance. This involves managing benefits like workforce development, resource efficiency, and social inclusion while avoiding harms such as automation without retraining and unsustainable resource use. ASM emphasises data-driven adaptability, allowing manufacturers to adjust operations dynamically for efficiency and resilience. It incorporates circular economy principles to support environmental sustainability and promotes a proactive, structured approach to technology adoption. In doing so, ASM helps firms align with Industry 5.0 objectives by combining operational performance with human and ecological priorities.

Designed as a practical and actionable vision, ASM addresses the reality of manufacturers navigating a competitive and disruptive business environment. It acknowledges the need to enhance productivity and financial outcomes while ensuring that digitalisation is optimised to avoid socio-environmental risks. ASM emphasises the strategic balancing of economic goals with sustainability, preventing negative externalities and protecting long-term viability. Accordingly, ASM is defined in this study as:

> a human-centric and sustainability-focused framework that integrates advanced technologies into manufacturing. It emphasizes responsible technology governance to balance efficiency with social well-being and resilience. ASM dynamically adapts processes to enhance productivity while systematically addressing the social and ecological values of Industry 5.0.

4.1. ASM vs. Existing paradigms

ASM bridges the gap between Industry 4.0's technology-led transformation, smart manufacturing practices, and Industry 5.0's sustainability goals. While these paradigms have shaped industrial production, none fully deliver balanced or practical sustainability outcomes.

Industry 4.0 represents a major industrial shift driven by advanced digital technologies. Initially focused on manufacturing digitalisation, it has expanded to sectors such as healthcare, tourism, and transportation (Sarbu 2022). Operating across micro and macro levels, it has reshaped value chains and enabled hyper-connected ecosystems (Maretto, Faccio, and Battini 2024). In manufacturing, Industry 4.0 has led to smart factories powered by real-time analytics and automation (Singh, Madaan, and Kumar 2023), forming a production model centred on productivity, efficiency, and optimisation (Mahmoodi et al. 2024). Despite these benefits, smart manufacturing has introduced serious sustainability concerns (Ghobakhloo et al. 2024; Grybauskas, Stefanini, and

Ghobakhloo 2022). As shown in Figure 2 and supported by SLR findings, these include increased energy demand, resource depletion, and e-waste. At the same time, skill gaps, job losses, and income inequality have revealed the social limitations of a purely technology-driven approach (Nadeem et al. 2024).

Industry 5.0 emerged to address these shortcomings by promoting a more human-centric and sustainable vision for industrial development (Maretto, Faccio, and Battini 2024). Rather than advancing technology for its own sake, it emphasises the governance of transformation and prioritises social and environmental values (Horvat, Jäger, and Lerch 2025). It aims to guide Industry 4.0's evolution through principles such as inclusiveness, resilience, and sustainability (Maddikunta et al. 2022; Xu et al. 2021). However, Industry 5.0 remains difficult to implement. Its wide scope, idealistic goals, and cross-sector ambitions often conflict with manufacturing realities (Mukherjee, Raj, and Aggarwal 2023). Implementation is hampered by fragmented ecosystems, uneven digital maturity, and the absence of practical guidance (Ghobakhloo et al. 2023b; Hein-Pensel et al. 2023).

ASM addresses this gap by offering a pragmatic evolution of smart manufacturing that embeds the goals of Industry 5.0 into the Industry 4.0 transformation process. It does not replace either paradigm but integrates their strengths while addressing their weaknesses. ASM retains the technological advantages of smart manufacturing while introducing strategic governance mechanisms to ensure alignment with sustainability, inclusion, and resilience. Table 1 summarises how ASM compares with other digital production models.

In addition to smart manufacturing, ASM can be compared with production frameworks such as cloud manufacturing, peer production, open production, and social manufacturing (SocialM). These models focus on decentralised networks, platform-based collaboration, and flexible value chains. Cloud manufacturing treats resources as on-demand services, while peer and open production allow distributed contributions via open platforms (Ellwein, Jainczyk, and Riedel 2023). SocialM extends this by creating cyber-physical-social ecosystems where resource communities provide manufacturing services to prosumers (Jiang, Ding, and Leng 2016). It supports community-driven production coordinated by social platforms (Shi et al. 2024). While SocialM restructures production systems, ASM governs digital transformation within existing setups. SocialM defines how production is organised (Guo and Jiang 2018), whereas ASM defines how transformation is managed to ensure sustainable and responsible outcomes.

These approaches are not mutually exclusive. Manufacturers adopting cloud manufacturing, peer production, or SocialM can apply ASM principles to ensure that digital transformation supports social responsibility and environmental goals. ASM draws from Industry 5.0's emphasis on human-centricity and equity (Sindhwani et al. 2022), complementing the collaborative nature of SocialM with a governance structure that safeguards workforce well-being, inclusion, and regulatory compliance.

Ultimately, ASM offers a cross-cutting governance framework applicable to diverse production contexts. By ensuring that digital transformation aligns with operational, social, and environmental objectives, ASM helps manufacturers achieve long-term sustainability and resilience.

5. ASM framework

Figure 3 presents the ASM framework, which outlines the development of a hyper-connected production ecosystem where technologies are integrated strategically to support sustainability. These technologies fall into two categories: foundational information and operations technologies (IT/OTs) and core emerging technologies.

Foundational IT/OTs are well-established in manufacturing and have evolved to meet sector needs. They form the base for ASM's development and include enterprise systems and computer-aided manufacturing tools (Nara et al. 2021). These technologies are essential for integrating Industry 5.0 innovations effectively (Huang et al. 2022; Maddikunta et al. 2022). Manufacturers pursuing ASM must already be proficient in deploying these systems. ASM also incorporates core emerging Industry 5.0 technologies to help manufacturers remain competitive. These include Cognitive Cyber-physical Production Systems (C-CPS) and intelligent automation tools (Gladysz et al. 2023). While these technologies can boost performance, they can also disrupt operations and create social challenges if not properly managed (Ghobakhloo et al. 2023b; Narkhede, Dohale, and Mahajan 2024b). Labelled 'emerging,' these tools are moving toward practical application, making their industrial impact increasingly tangible.

The ASM framework highlights that technology can have both positive and negative effects on sustainability. Based on dynamic capability theory (Teece 2014), ASM emphasises the need to integrate, build, and reconfigure capabilities to adapt to rapid technological change while maintaining resilience. Resilience is defined as the ability to anticipate, absorb, and recover from disruptions

Table 1. Comparative Analysis of ASM with existing frameworks.

Aspect	Industry 4.0	Smart manufacturing	Industry 5.0	ASM
Core Focus	Full-scale digital transformation of industries through emerging disruptive technologies (e.g. AI) to enable hyper-connected value chains and business ecosystems across sectors.	Technology-driven optimisation of manufacturing processes within the factory, focusing on operational efficiency, productivity, and quality enhancement.	Human-centric and sustainability-driven governance framework aimed at regulating Industry 4.0 transformation while ensuring resilience, well-being, and inclusivity.	Balanced transformation integrating Industry 4.0's technological advancement, smart manufacturing's operational focus, and Industry 5.0's sustainability goals that are operationally feasible.
Scope of Application	Broad, cross-industry application beyond manufacturing, including healthcare, transportation, and tourism, with impacts across entire value networks.	Narrower in scope, focused primarily on enhancing production processes within the four walls of the factory rather than across business networks.	Cross-industry governance framework addressing sustainability challenges and promoting human-centric practices across industrial ecosystems.	Primarily applied to manufacturing and the operations level but expanding to stakeholder involvement, ensuring that manufacturing digital transformation remains inclusive, sustainable, and socially responsible.
Environmental Sustainability Focus	Efficiency-driven with incidental sustainability gains; often overlooks environmental and social impacts like energy consumption, resource depletion, and e-waste.	Primarily focuses on operational efficiency, often lacking explicit sustainability goals or mechanisms to address broader ecological and social challenges.	Promotes environmental sustainability and resilience by idealistically pursuing inclusive sustainability goals such as renewable energy, circular economy practices, and sustainable production ecosystems.	Embeds sustainability into manufacturing processes through adaptive governance, ensuring continuous alignment with evolving environmental standards, but in an economically viable manner.
Human-Centric Approach	Technology-first approach, often leading to job displacement, skill mismatches, and reduced worker autonomy due to automation-driven production models.	Enhances worker safety and productivity within factories but lacks structured approaches to workforce well-being, skill development, and socio-economic inclusion.	Emphasises worker empowerment, skill development, and collaborative environments, requiring that technology enhances rather than replaces human roles.	Promotes workforce empowerment, participatory decision-making, and equitable access to technological advancements through adaptive governance structures.
Governance Approach	Lacks structured sustainability governance, leading to fragmented implementations and unintended social and environmental consequences.	Focuses on operational control and performance monitoring within manufacturing plants, without addressing broader sustainability or social responsibility frameworks.	Promotes holistic governance but struggles with practical implementation due to its broad scope, fragmented ecosystems, and cross-sector complexity.	Implements firm and supply chain-level adaptive governance frameworks, ensuring manufacturing practices align with evolving sustainability standards, stakeholder expectations, and regulatory requirements.
Operational Feasibility	High feasibility for technology adoption but often fragmented due to coordination complexity across value chains. Higher operationalise success rate among larger and leading firms.	Feasible for factory-level implementation but constrained by its inward focus, lacking mechanisms to extend benefits across supply chains and ecosystems.	Ambitious but challenging to implement practically due to its broad, cross-industry goals and lack of sector-specific operational models.	Feasible within manufacturing, offering sector-specific governance mechanisms that balance productivity, sustainability, and social inclusion.
Unique Value Proposition	Enhances productivity and efficiency via advanced automation and digitalisation, creating hyper-connected industrial ecosystems and offers unique competitiveness and monopolisation opportunities.	Improves production efficiency, quality, and productivity within the factory but without holistic governance to purposefully address environmental or societal impacts.	Promotes human-centric and resilience-driven industrial transformation but lacks sector-specific operational models for practical implementation or rigid economic feasibility for implementation.	Integrates the technological strengths of Industry 4.0 and smart manufacturing with Industry 5.0's sustainability ideals (those practical and feasible), providing a framework for balanced, adaptive manufacturing.

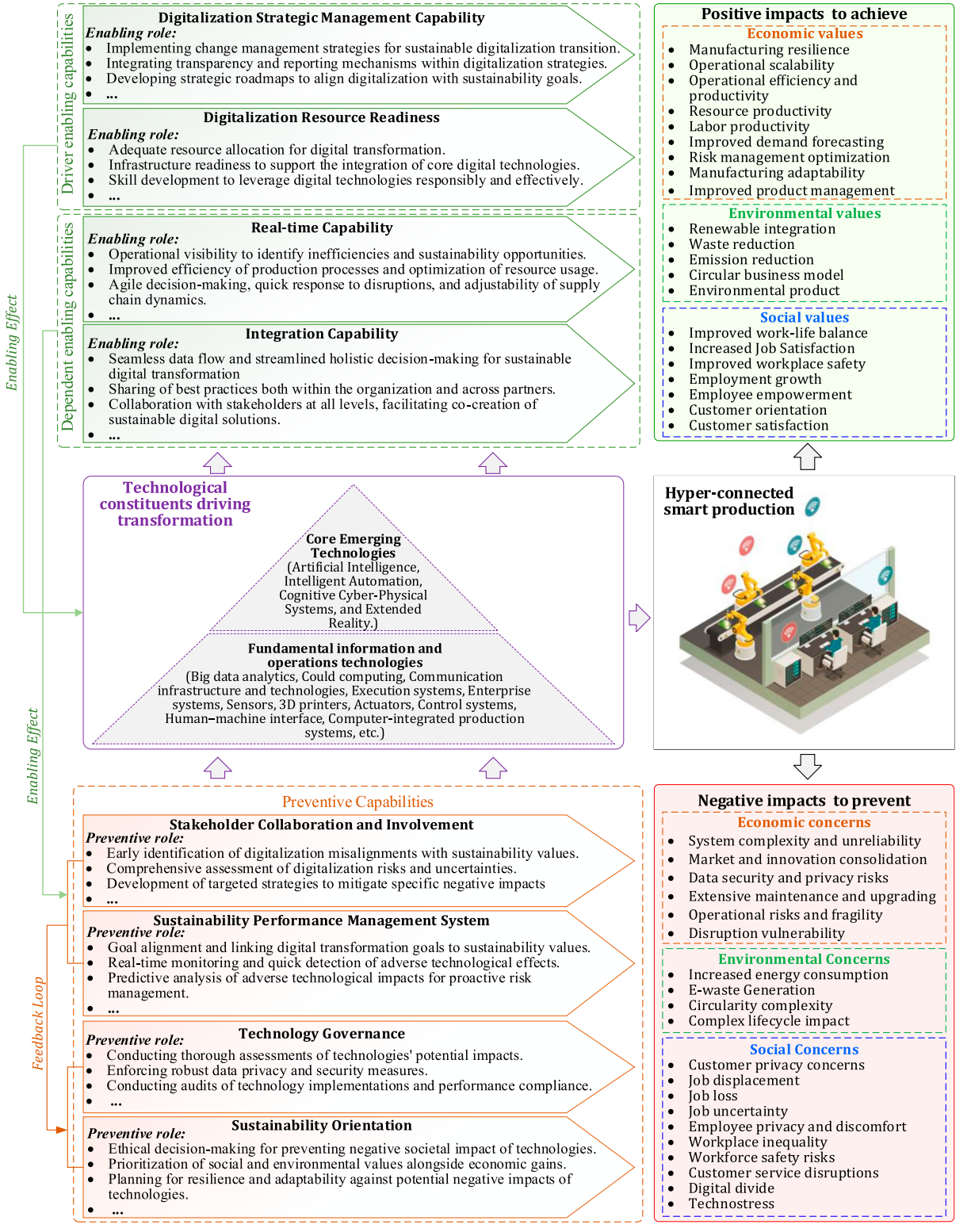
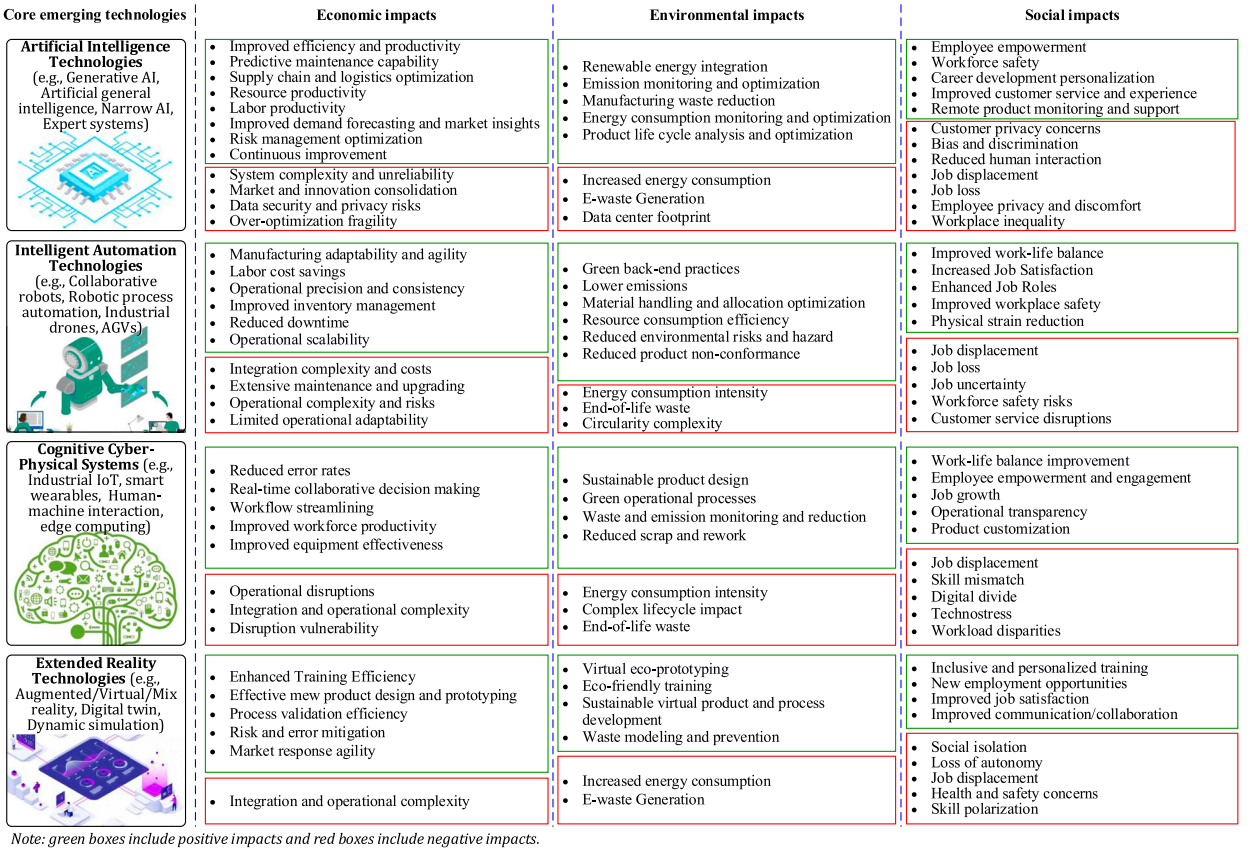


Figure 3. The proposed ASM framework.

without compromising long-term objectives (Sindhwani et al. 2022). In Industry 5.0, which prioritises social and environmental goals, resilience is critical for sustaining innovation and continuity (Leng et al. 2023a). Without it, digitalisation may lead to fragmented systems, social inequality, or environmental harm (Leng et al. 2022).

ASM operationalises this framework through two types of organisational capabilities. First, 'enabling capa-



Note: green boxes include positive impacts and red boxes include negative impacts.

Figure 4. Core ASM technologies and their potential impacts on sustainability.

bilities' enhance the positive effects of technology. These include strategic digitalisation management and resource readiness, supported by real-time and integration capabilities. Together, they help firms identify and scale digital innovations to drive both productivity and environmental performance, aligning with dynamic capability pathways. Second, 'preventive capabilities' mitigate technology risks. Based on risk governance frameworks (e.g. IRGC 2017), these include stakeholder collaboration, technology governance, and sustainability performance management. These mechanisms help organisations adjust strategies to avoid social, environmental, or workplace harm, ensuring digitalisation remains ethical and inclusive.

The framework's components are interconnected. Each technology can affect sustainability differently, and dependencies often exist among enablers and preventers. These relationships are detailed in the following sections.

5.1. Technological constituents

As Figure 4 implies, the technological drivers of ASM entail foundational IT/OTs and core emerging technologies. Foundational IT/OTs are well discussed within the literature (e.g. Maddikunta et al. 2022), whereas core emerging technologies of ASM are less elaborated. The content analysis results imply that the core technologies of ASM under the Industry 5.0 context can be broadly structured into *four* categories. These categories and the positive and negative impacts they can potentially imply for sustainability values are systematically presented in Figure 4 and discussed in the following.

AI technologies are ASM's most progressive technological component as they reshape industries and human interactions (Mo et al. 2023; Passalacqua et al. 2025). AI's pervasive impact is evident in manufacturing, healthcare, finance, and entertainment. These developments have sparked debates on regulations and utilisation, highlighting concerns about ethics, privacy, and job displacement (Rana et al. 2022). AI's capabilities in predictive analytics, natural language processing, and machine learning raise vital questions about accountability and transparency (Gupta et al. 2023). Amid these conversations, AI's potential for innovation, efficiency, and societal implications remains at the forefront of discussions, underscoring its central role in shaping our current and future landscapes (Chang and Ke 2024). AI application in manufacturers is associated with transformational impacts on various sustainability facets (Plathottam et al. 2023). From an economic standpoint, AI boosts predictive maintenance capabilities that optimise machinery

reliability and minimise costly downtimes (Psarommatis, May, and Azamfirei 2023). Supply chain and logistics gain efficiency through AI-driven optimizations, contributing to reduced waste and resource depletion (Pournader et al. 2021). Workforce productivity is propelled as AI streamlines resource allocation and fosters labour efficiency (Sun and Song 2025). Moreover, the accurate demand forecasting enabled by AI enhances market insights and fosters proactive risk management strategies (Sohrabpour et al. 2021). However, there are potential caveats. The complex nature of AI systems can lead to unreliability, and market consolidation by AI-centric firms might stifle competition and innovation (Choi et al. 2022). Moreover, concerns arise over data security and privacy in the era of heightened data dependence, while the pursuit of optimisation may render systems fragile in dynamic environments (Ahmad et al. 2021).

Under environmental sustainability, manufacturers can pivot towards sustainable energy sources thanks to AI-facilitated renewable energy integration (Ahmad et al. 2022). Real-time emissions monitoring and reduction can become feasible, fostering eco-friendly practices. AI's role in waste reduction within manufacturing processes contributes to ecological well-being (Ching et al. 2022). Energy consumption monitoring and optimisation ensure efficient resource use, and AI-driven life cycle analyses can offer informed decisions for sustainable products (Ghobakhloo and Fathi 2021). However, the energy-intensive nature of AI operations poses the risk of increased energy consumption, and the growth of electronic waste is an environmental concern (Kunkel and Tyfield 2021). Additionally, AI's computational requirements can heighten data centre energy footprints, warranting careful consideration in pursuing holistic sustainability (Nishant, Kennedy, and Corbett 2020).

In the social dimension, AI's influence is profound, empowering employees through data-driven decision-making and enhancing workplace safety. AI can offer carrier development personalisation, bolstering workforce development. It can also significantly improve customer interactions and experiences (Vrontis et al. 2022). Nevertheless, AI raises numerous societal concerns, including privacy concerns, spotlighting the need for customer data protection (Du and Xie 2021). The potential for AI algorithms to perpetuate bias and discrimination raises ethical questions. Furthermore, the increased reliance on AI threatens to diminish human interaction, leading to potential job displacement and loss, thereby aggravating workplace inequalities (Grybauskas, Stefanini, and Ghobakhloo 2022). In the ASM framework, harnessing the positive potential of AI is contingent on understanding and addressing these multifaceted impacts to achieve sustainable outcomes.

While AI presents transformative opportunities across sustainability dimensions, its impact may vary significantly based on contextual factors, particularly between developed and developing economies (Rashid et al. 2025). In developed economies, where advanced digital infrastructure, skilled workforces, and mature regulatory frameworks are established, AI adoption tends to be swifter and more impactful, particularly in areas like predictive maintenance, supply chain optimisation, and energy efficiency with measurable sustainability gains. However, in developing economies, while infrastructure gaps and resource constraints may present initial challenges, they also create unique opportunities for leapfrogging (Kar, Choudhary, and Singh 2022; Kulkarni, Joseph, and Patil 2024). Without legacy systems to overhaul, many manufacturers in developing regions can adopt cutting-edge AI-driven solutions from the outset, bypassing incremental upgrades common in mature markets. Cloud-based AI platforms can enable real-time monitoring and predictive analytics without significant on-premise infrastructure investments, while mobile-driven AI applications can enhance supply chain visibility in remote areas (Anshari et al. 2025). Additionally, the growing focus on renewable energy in developing regions offers synergistic opportunities, as AI-enabled energy optimisation can accelerate the transition to sustainable power sources (Rashid et al. 2025). Nevertheless, regulatory frameworks and data governance structures may still require strengthening to ensure ethical AI adoption, while the energy demands of AI infrastructure could strain local power grids if not carefully managed.

Intelligent automation technologies integrate robotics and process automation to enhance and streamline various aspects of industrial operations and processes (Ghobakhloo et al. 2023a). It uses smart systems and algorithms to automate tasks and make data-driven decisions, improving efficiency, accuracy, and adaptability. Intelligent automation combines the capabilities of machines and software to perform tasks that traditionally require human intervention (Ng et al. 2021). As a result, intelligent automation technologies have emerged as transformative forces in the manufacturing landscape. These technologies are redefining how industries operate, with implications for both economic and sustainability outcomes (Lu, Xu, and Wang 2020).

From an economic perspective, implementing these technologies brings about several positive impacts. Manufacturing can gain enhanced adaptability and agility

as automation allows swift reconfiguration of production lines (Ghobakhloo et al. 2023a). Labour cost savings can also be achieved through reduced manual intervention and increased operational efficiency (Wirtz, Kunz, and Paluch 2021). These technologies can further boost precision and consistency in operations, improving product quality while enhancing inventory management and optimising resource allocation (Wang et al. 2021). These technologies can also streamline processes and reduce downtime and operational scalability (Ng et al. 2021). However, the integration complexity and costs associated with implementing these technologies can pose challenges, along with the need for extensive maintenance and upgrading to ensure sustained performance (Vrontis et al. 2022). Manufacturers should effectively manage these technologies' operational complexity, associated risks, and limited adaptability in specific scenarios to prevent negative productivity impacts.

Intelligent automation technologies can offer significant benefits for environmental sustainability. These technologies can facilitate green backend practices by minimising paper-based workflows and reducing the need for physical resources (Nishant, Kennedy, and Corbett 2020). They can lower emissions through optimised operations and reduced reliance on energy-intensive processes (Ghobakhloo and Fathi 2021). Intelligent automation can also lead to efficient resource consumption by optimising material handling and allocation. This can reduce environmental risks and hazards associated with conventional manufacturing processes while contributing to reduced instances of product non-conformance (Ching et al. 2022). Nevertheless, the increased energy consumption intensity during automation operations, the management of end-of-life waste from automated components, and the complexity of achieving circularity in automation components require careful consideration (Jones, Hutcheson, and Camba 2021).

On the social front, intelligent automation technologies present a mixed impact. Positive effects include improved work-life balance and increased job satisfaction for employees who can focus on higher-value tasks (Braganza et al. 2022). Enhanced job roles and responsibilities enhance skill enrichment and professional growth (Vrontis et al. 2022). Workplace safety is augmented through reduced exposure to hazardous tasks (Liu et al. 2020). However, potential adverse social effects include job displacement and uncertainty due to the automation of specific tasks, which can impact workforce dynamics (Fierro, Caiani, and Russo 2022; Ghobakhloo et al. 2024). The risk of job loss and customer service disruptions poses challenges for a smooth transition to automated processes (Chiarini et al. 2023). Balancing these social effects requires strategic planning and a supportive transition for the workforce.

C-CPS represents technological innovations that merge the physical and digital realms of manufacturing. These technologies form a cohesive ecosystem where sensors, devices, and humans collaboratively interact to enhance industrial operations (Ghobakhloo et al. 2023c). Their positive economic effects are significant. C-CPS technologies can allow manufacturers to achieve higher precision in their processes by reducing error rates through data-driven insights (Yao et al. 2019). Real-time collaborative decision-making facilitated by C-CPS will enable businesses to respond swiftly to evolving situations. Workflow streamlining, driven by the seamless integration of various devices and systems under C-CPS, can optimise operational efficiency (Wang et al. 2022). Moreover, improved workforce productivity can be achieved through smart wearables and human-machine interaction, where real-time insights aid employees in making informed decisions (Patel et al. 2022). Additionally, these technologies enhance equipment effectiveness, as real-time monitoring and analysis enable predictive maintenance and performance optimisation. However, implementing C-CPS is challenging (Yao et al. 2019). Operational disruptions can occur due to technical glitches or malfunctioning sensors, potentially halting production processes. Integrating and managing the complexity of various C-CPS components can be intricate and costly (Wankhede and Vinodh 2022). This complexity also leaves manufacturers vulnerable to disruptions caused by cyber threats.

On the environmental front, C-CPS offers positive outcomes by promoting sustainable practices. Through sustainable product design and green operational processes, integrating IIoT and edge computing can lead to energy savings and reduced waste generation (Leiden, Herrmann, and Thiede 2021). Real-time monitoring and analysis enable waste and emission reduction, contributing to environmental conservation (Kamble, Gunasekaran, and Gawankar 2018). Nonetheless, there are potential negative impacts to consider. The increased energy consumption intensity, particularly in data processing, might offset the benefits of energy efficiency (Lu, Xu, and Wang 2020). The complex lifecycle impact of integrating and disposing of C-CPS components also poses environmental challenges. Moreover, disposing of outdated equipment could contribute to electronic waste (Thiede 2018).

On a societal level, C-CPS can provide valuable positive effects, such as improving work-life balance for employees through automation and remote access or empowering the workforce with cognitive technologies to enhance engagement and morale further (Pinzone

et al. 2020). As new roles emerge due to the demand for technology-savvy professionals, job growth can be fostered (Waschull et al. 2020). C-CPS can boost operational transparency as real-time insights are shared, contributing to a culture of trust and collaboration. C-CPS further enables product customisation, enhancing customer experience (Leiden, Herrmann, and Thiede 2021). Despite these benefits, potential societal challenges are evident. Job displacement due to automation and the demand for new skill sets may lead to skill mismatch among the workforce (Waschull et al. 2020). Unequal access to technology can exacerbate the digital divide, creating disparities in technological adoption (Ghobakhloo et al. 2023a). Overreliance on technology might also lead to technostress and burnout among employees. Moreover, uneven technology adoption can develop imbalances in workloads and job roles (Zolg, Heiden, and Herbig 2021). As manufacturers navigate the integration of CCPS, careful consideration of these impacts is crucial for achieving a harmonious balance between technological advancement and sustainable societal outcomes.

Extended Reality (XR) technologies, such as augmented reality, virtual reality, mixed reality, digital twin, and dynamic simulation, can redefine how manufacturers envision, prototype, and validate their products and processes (Ching et al. 2022). These technologies also offer promising economic benefits, starting with enhanced training efficiency through realistic simulations that empower workers with practical skills (de Giorgio et al. 2023). They facilitate effective new product design and prototyping, enabling rapid iterations and reducing development cycles (Arrighi and Mougenot 2019). Process validation becomes efficient through virtual tests, minimising errors and saving time. Risk mitigation can also be improved with data-driven insights, and market response agility is boosted with real-time analytics, aiding decision-making (Cardenas-Robledo et al. 2022).

However, integration and operational complexity pose a considerable economic challenge, as harmonising XR technologies with existing processes requires meticulous planning and resources (Masood and Egger 2019). While XR technologies hold environmental promise, such as virtual eco-prototyping that reduces physical waste and eco-friendly training, they also contribute to increased energy consumption due to processing demands and potential electronic waste generation (Perno, Hvam, and Haug 2022). From the social value perspective, XR technologies can offer inclusive and personalised training experiences that transcend traditional barriers. They generate new employment opportunities in XR content creation and management, enhancing job markets (de Giorgio et al. 2023). Moreover, they can improve job satisfaction by creating engaging and dynamic work experiences. Thanks to XR technologies, enhanced communication and collaboration can foster efficient teamwork, improving the work environment (Marino et al. 2021). Nonetheless, XR technologies can lead to negative social effects. Prolonged use might contribute to social isolation as physical interactions diminish. The loss of autonomy might surface as tasks shift to XR systems (Merkx and Nawijn 2021). Job displacement can occur as automation becomes prevalent, and concerns about health and safety risks in XR-rich environments might arise. Skill polarisation might also emerge, where those adept in XR technologies flourish while others face challenges (Sung 2018).

The findings in Figure 4 show a clear trend. The potential economic benefits of emerging technologies often outweigh their negative economic impacts. This is even more evident in their environmental implications. Economic downsides are usually linked to the complexity of these technologies and their vulnerability to disruption, which can lead to failed digital initiatives. Environmental drawbacks mainly relate to high energy use and end-of-life disposal issues. However, the picture is different when it comes to social impacts. As shown in Figure 4, the potential negative social effects are often as significant as, or greater than, the positive ones. Although these technologies are expected to support the human-centric goals of Industry 5.0, they can harm social well-being if not properly managed. This highlights the need for strong governance to ensure that their development and use support, rather than undermine, societal values.

5.2. Preventing capabilities

As shown in Figure 3, several critical mechanisms help prevent the negative sustainability impacts associated with ASM's core technologies. These preventers are essential for ensuring that technological advancement remains aligned with environmental and societal well-being.

The first preventer is *stakeholder collaboration and involvement*, which plays a key role in aligning digitalisation efforts with sustainability objectives. Engaging various stakeholders helps identify, evaluate, and manage potential sustainability risks that may arise from digital transformation (Ghobakhloo et al. 2023b). Early identification of misalignments between digital initiatives and sustainability values is critical, particularly as organisations adopt advanced technologies at speed. Employees, communities, regulators, and consumers contribute essential insights that reveal gaps between the intended outcomes of digitalisation and its real-world

effects (Hellemans, Porter, and Diriker 2022). These inputs enable responsive and informed decision-making throughout the transformation process (Ghobakhloo et al. 2023b).

Stakeholder involvement also supports proactive assessment of digitalisation-related risks and uncertainties (Guo et al. 2023). Potential risks include ethical concerns, cybersecurity vulnerabilities, and societal disruptions (Chiarini et al. 2023). Stakeholders bring diverse perspectives. For example, employees may focus on job security while local communities may highlight ecological risks (Hellemans, Porter, and Diriker 2022). This diversity of viewpoints strengthens manufacturers' ability to anticipate challenges and adjust strategies accordingly, reducing the likelihood of negative outcomes.

Collaboration also enables the co-development of targeted strategies to address specific concerns. Preventing adverse effects depends on integrating stakeholder knowledge into digital decision-making (Longo, Padovano, and Umbrello 2020). Through engagement, companies can create tailored responses. For instance, if stakeholders raise concerns about job displacement, training and reskilling initiatives can be introduced. If environmental risks are highlighted, sustainable design practices or green technology alternatives can be prioritised (Ching et al. 2022). This approach ensures that responses are both timely and relevant.

The second preventer is the *sustainability performance management system (SPMS)*, which helps ensure that digital transformation stays aligned with sustainability values. While technological progress in Industry 5.0 can drive rapid gains, it also risks overemphasising short-term efficiency or economic returns (Renda et al. 2022). SPMS frameworks help maintain balance by embedding sustainability targets, such as emissions reduction or social equity, into digitalisation plans (Cillo et al. 2022). This alignment safeguards against the erosion of social or environmental performance in pursuit of economic growth. SPMS also supports real-time detection of negative effects. Complex technologies can produce unintended sustainability consequences, such as increased energy use from automation systems (Wang et al. 2023b). A well-designed SPMS uses data analytics to continuously track digital performance and identify such issues early. This enables timely corrective action and helps maintain consistent alignment with sustainability goals (Qian et al. 2020). In addition, SPMS can use predictive analytics to anticipate longer-term risks. While real-time monitoring captures present deviations (Qiao, Liu, and Ma 2021), predictive functions analyse patterns and project possible future outcomes. For example, automation trends might signal future workforce displacement. With this insight, manufacturers can adopt measures to address social impacts before they materialise (Nishant, Kennedy, and Corbett 2020; Plathottam et al. 2023).

The third preventer is *technology governance*, which refers to how firms manage and oversee their digital systems to ensure alignment with sustainability and ethical standards (Elazhary et al. 2023). Governance frameworks support impact assessments before widespread adoption. Since emerging technologies can fundamentally reshape operations, governance policies must evaluate not only expected economic returns but also potential environmental and social impacts (Ghobakhloo et al. 2023b; Guo et al. 2023). For example, before implementing an automation solution, it is essential to assess its energy demands, compatibility with renewables, and implications for the workforce (Bai et al. 2020). Such a structured evaluation ensures that technology adoption supports long-term sustainability objectives.

Another important function of technology governance is protecting data privacy and security. Data plays a central role in Industry 5.0, and the protection of personal, organisational, and process data is critical (Huang et al. 2022; Sharma, Yadav, and Chopra 2020). Weak security can lead to data breaches that erode customer trust and cause financial losses (Culot et al. 2019). Governance systems should include formal security protocols, encryption, access control, periodic audits, and strict adherence to data protection laws.

Technology audits also serve as a proactive governance tool. The integration of cognitive and digital technologies can be disruptive, and audits help ensure these systems operate as intended and meet defined sustainability and performance benchmarks (Ghobakhloo et al. 2023b; Grybauskas, Stefanini, and Ghobakhloo 2022). For instance, audits of IIoT deployments can evaluate connectivity, data integrity, and operational efficiency. Where gaps are found, corrective actions can be taken to minimise losses. Audits also reinforce regulatory compliance and accountability (Alkaraan et al. 2023), providing a structured mechanism to prevent adverse impacts.

The fourth preventer is *sustainability orientation*, a mindset that promotes ethical, inclusive, and long-term thinking across all levels of digital transformation (Jagani and Hong 2022). This orientation embeds ethical considerations into decision-making processes, encouraging manufacturers to evaluate the societal implications of AI, data use, and human-machine interaction (Salamzadeh et al. 2022). Sustainability orientation ensures that environmental and social values are treated as equal to financial objectives. Organisations are encouraged to design digital transformations that minimise ecological damage and protect social cohesion (Dabi#263; et al. 2023).

A sustainability-oriented approach also promotes resilience and adaptability. It recognises that digital transformation can introduce volatility and that firms must prepare for disruption (Forliano et al. 2023). Proactive strategies might include reskilling programmes to manage job displacement or investing in infrastructure that enhances organisational flexibility. Digital tools such as AI-powered decision systems, predictive analytics, and digital twins further support resilience. These tools can detect threats early and enable agile, informed responses to change (Leng et al. 2023b). With strategic use, firms can anticipate disruptions and reduce their impact on operations and social structures (DiBella et al. 2023; Ghobakhloo et al. 2023b).

Together, these four preventers form a comprehensive framework to prevent the negative consequences of digital transformation in ASM. Their effectiveness depends on proactive implementation and integration into the strategic and operational layers of manufacturing. By embedding these mechanisms into their transformation processes, firms can achieve technological advancement without compromising social or environmental sustainability.

5.3. Enabling capabilities

The enabling capabilities shown in Figure 3 are essential for realising the positive sustainability outcomes of ASM's core technologies. They play a key role in capturing and directing these benefits toward sustainability goals. When effectively managed, they accelerate and reinforce the conversion of technological potential into actual sustainability value.

Digitalization Strategic Management Capability (DSMC) supports organisations in navigating digital transformation by providing a clear framework for action (Ching et al. 2022). This capability consists of essential competencies and practices for addressing both challenges and opportunities in digitalisation (Castelo-Branco et al. 2022). One important DSMC function involves change management strategies that support the workforce during the shift toward digital and sustainable manufacturing (Pozzi, Rossi, and Secchi 2023). These strategies help prepare employees for sustainable practices, address job displacement and skill mismatches, and reduce technostress (Piccarozzi et al. 2022).

DSMC's second function involves building transparency and reporting mechanisms that improve accountability through open communication and data sharing (Ghobakhloo et al. 2023b). These mechanisms allow manufacturers to share progress, challenges, and achievements with stakeholders and encourage deeper engagement (Piccarozzi et al. 2022; Pozzi, Rossi, and Secchi 2023). The third DSMC function is developing strategic roadmaps that align digital transformation with sustainability targets (Ching et al. 2022). These roadmaps help guide the adoption of new technologies while identifying and managing social and environmental risks (Singh, Madaan, and Kumar 2023). By aligning digitalisation with sustainability from the outset, DSMC ensures a responsible transformation that avoids conflict between innovation and long-term goals (Narayanan 2023).

Digitalization Resource Readiness (DRR) is also critical, as it ensures that ASM's technologies are supported with the right resources, particularly funding (Srivastava et al. 2022). Adequate financial support is necessary to acquire and implement digital tools and build smart factories with real-time connectivity that improve efficiency and resource use (Mukherjee, Raj, and Aggarwal 2023; Xu et al. 2021). DRR also helps ensure that green technology initiatives receive the financial backing they need to support sustainability goals (Wang et al. 2023a).

Another part of DRR is infrastructure readiness, which is necessary for integrating ASM's digital technologies. A mature technological infrastructure enables efficient data flow that supports automation and informed decision-making (Leng et al. 2022; Psarommatis, May, and Azamfirei 2023). Real-time process monitoring becomes possible, helping to reduce waste, prevent data breaches or downtime, and strengthen environmental and social outcomes (Hein-Pensel et al. 2023; Xian et al. 2024). DRR also includes workforce development. Well-trained employees are better equipped to use technology responsibly, helping avoid negative outcomes such as privacy violations or algorithmic bias (Ghobakhloo et al. 2023a). At the same time, they can maximise the benefits of digital tools to improve productivity and reduce waste (Longo, Padovano, and Umbrello 2020).

Real-time capability refers to the ability to collect, analyse, and respond to data as it is generated. This involves using sensors, analytics, and communication technologies to support real-time monitoring, control, and decision-making (Ghaleb, Zolfagharinia, and Taghipour 2020). This capability strengthens ASM's sustainability by improving visibility and enabling fast identification of inefficiencies and opportunities (Ching et al. 2022; Qian et al. 2020). For example, manufacturers can detect problems like bottlenecks or material waste as they happen and immediately take corrective action (Qiao, Liu, and Ma 2021; Singh, Madaan, and Kumar 2023). It also allows rapid responses to energy inefficiencies and spikes in emissions (Darwish, El-Wakad, and Farag 2021).

Real-time capability improves system coordination and supports continuous optimisation of energy use, material flow, and costs (Ahmad et al. 2022; Ardanza

et al. 2019). It also supports quick decision-making when disruptions occur, allowing supply chains to adapt to changes in demand or supply (Fernandez-Viagas and Framinan 2022). Real-time insights enable manufacturers to dynamically adjust procurement, production, and delivery to reduce inefficiencies and environmental impacts (Ardanza et al. 2019; Hofmann et al. 2019).

Integration capability refers to a manufacturer's ability to connect and use data across internal departments and external partners (Enrique et al. 2022). It forms the backbone of ASM by enabling interoperability between different systems, technologies, and organisations. Vertical integration connects operational data from devices and control systems to higher-level platforms such as ERP systems and dashboards (Jaskó et al. 2020). This ensures key performance data are not only collected but used for strategic decisions and ongoing improvements (Maddikunta et al. 2022; Wang et al. 2023a). Achieving vertical integration often depends on middleware, industrial IoT platforms, and standard communication protocols that link older systems with modern enterprise tools (Mantravadi, Møller, and Schnyder 2022).

Horizontally, integration capability connects manufacturers with their partners, logistics providers, regulators, and customers. This kind of connectivity is especially important in digital ecosystems that require transparency and coordinated sustainability efforts (Ghobakhloo et al. 2022). Horizontal integration requires secure APIs, cloud platforms, and open data standards for joint planning, shared indicators, and collaborative reporting (Ching et al. 2022). Blockchain can also help by creating trust, auditability, and traceability in transactions and data sharing across networks (Alkaraan et al. 2023).

ASM recognises that many firms still use legacy systems that are not designed for easy integration. Instead of replacing these systems, integration capability includes ways to augment them through edge computing, data virtualisation, and API layers. This allows legacy technologies to operate within a modern, connected environment, making ASM more inclusive and scalable (Bustamante et al. 2023; Mantravadi, Srai, and Møller 2023). Beyond technology, integration capability also supports collaboration both within and beyond the organisation. Internally, it helps share best practices, coordinate across teams, and centralise sustainability efforts (Ghobakhloo et al. 2022). Externally, it enables co-development of sustainable innovations with suppliers, customers, and authorities (Abdel-Basset, Mohamed, and Chang 2025). Examples include using supplier platforms to exchange performance data, providing customers with product lifecycle details, or working with regulators on streamlined compliance reporting (Dwivedi et al. 2023). Local community involvement can also support talent development and environmental initiatives (Renda et al. 2022), reinforcing ASM's focus on human-centric value.

In short, integration capability is more than a technical requirement. It is a strategic enabler of intelligence, innovation, and adaptive sustainability governance. By making sure that systems and stakeholders can connect and collaborate regardless of their digital maturity, this capability supports the comprehensive transformation of manufacturing ecosystems envisioned by Industry 5.0.

6. The case study application

The case study focuses on InnMedTech (anonymized name), a medium-sized healthcare product manufacturer with an estimated €48 million annual turnover. The company specialises in customisable medical devices, including wearable health monitors, smart prosthetics, and orthopedic supports. Operating under a hybrid B2B and B2C model, the company collaborates with healthcare providers while also selling directly to consumers through an online platform, enabling users to configure devices tailored to their individual needs.

By 2019, increasing demand for personalised healthcare solutions fuelled InnMedTech's rapid growth. However, as the company scaled, operational inefficiencies emerged, jeopardising its growth trajectory and expansion plans. The complexity of customised production, combined with fragmented digital workflows and manual processes, resulted in longer lead times, rising operational costs, and inconsistent product quality. These challenges were further exacerbated by inefficient decision-making across functions, material waste, and workforce stress due to insufficient digital support.

Recognising the urgent need for transformation, InnMedTech initiated a digitalisation strategy aligned with Industry 4.0 principles, aiming to enhance production efficiency, product quality, decision-making efficiency, and supply chain integration through digitalisation and modern process technologies. However, it quickly became evident that a standard Industry 4.0 transformation may fall short. Given its business model, built on a sustainability-driven value proposition, alongside regulatory pressures for socio-environmentally responsible manufacturing in the healthcare sector, the company required a more balanced and adaptive approach to digital transformation.

To address these challenges, InnMedTech adopted the ASM framework, leveraging its role as a partner in the corresponding EU-funded project aimed at developing sustainable digitalisation strategies. The case company applied the ASM framework through a structured, staged

approach to its digitalisation programme, ensuring transformation efforts were both strategic and sustainable. This stage-gated implementation followed the sequential adoption of ASM principles, starting with DSMC and DRR, which established the leadership, infrastructure, and competencies necessary for transformation. Once the foundation was in place, Stakeholder Collaboration, Technology Governance, and SPMS were introduced to guide responsible technology adoption and align digitalisation efforts with sustainability goals. Importantly, the ASM framework positioned this initiative not as a one-off project but as an ongoing programme, reflecting the ASM philosophy that sustainable digital transformation is continuous and evolutionary. As the company progressed, many ASM principles (core capabilities) persisted and evolved, with periodic updates driven by performance audits, stakeholder feedback, and technological advancements. While the principles were introduced sequentially, their contributions quickly became parallel, creating an ecosystem where advancements in one area reinforced progress in others. For instance, the implementation of real-time capability not only improved operational visibility but also enhanced sustainability tracking, stakeholder engagement, and system integration, transforming the digitalisation process into a dynamic, interconnected system. This continuous programme structure ensured that ASM principles remained adaptable and responsive to the company's evolving needs.

Stage one: Strategic management capability and resource readiness (DSMC and DRR)

At this stage, the company conducted a comprehensive technology transformation readiness assessment to evaluate its existing technological and resource capabilities. Table 2 presents a summary of the assessment findings, highlighting key dimensions of readiness, development priorities, and resource allocation requirements. The readiness assessment in Table 2 offers a clear view of InnMedTech's digital transformation readiness, highlighting strengths, gaps, and strategic priorities. This evaluation provided a phased approach for advancing the company's digital transformation readiness under the ASM framework, ensuring that foundational capabilities are established before technology-driven outcomes are pursued.

Table 2 implies that two dimensions, Leadership Support and Collaboration Culture, were prioritised as the highest development areas, both rated at priority level 1, despite moderate strategic importance. This reflects their foundational role in driving transformation. Leadership was recognised as critical for setting strategic direction, resource allocation, and cross-functional alignment. Without strong leadership, technology adoption efforts risk fragmentation and misalignment. Collaboration culture was identified as equally essential for fostering teamwork, knowledge sharing, and stakeholder engagement, ensuring the success of technology-driven initiatives in the case company. Both dimensions, though not the most resource-intensive, were identified as pivotal in creating the organisational conditions for sustainable transformation.

Employee Competence and Strategic Alignment were assigned priority level 2, reflecting their enabling role in supporting technology adoption while aligning digital initiatives with business goals. Workforce readiness underscored the need for targeted upskilling to ensure employees can effectively operate advanced systems. Strategic alignment emphasised the importance of integrating digital initiatives with InnMedTech's sustainability goals and regulatory requirements. This dimension required the highest resource allocation intensity, signalling the complexity and investment needed to align transformation with long-term objectives.

Smart Factory Technologies, despite being the most strategically important dimension, was assigned priority level 3, reflecting the need to first strengthen leadership, workforce competence, and strategic alignment. Its low readiness level highlights InnMedTech's limited deployment of advanced manufacturing systems, while the resource allocation intensity indicates substantial investment requirements for technology acquisition, system integration, and process restructuring.

Smart Operations and Value Network Integration, both rated at priority level 4, depend on upstream readiness factors such as leadership, workforce competence, and smart factory implementation. Smart operations with moderate readiness level and resource allocation intensity, highlighted the need for real-time production monitoring and adaptive planning. Value network integration emphasised the importance of seamless digital connectivity with supply chain partners, healthcare providers, service providers, and customers, without which collaborative production planning and synchronised operations would remain unattainable in the case company.

At the lowest development priority of 5, Data-Driven Services and Smart Products, despite high strategic importance scores, were viewed as outcomes rather than enablers of transformation. The resource allocation intensities for data-driven services and smart products reflect moderate investment needs, reinforcing their position as secondary priorities once the company achieves operational maturity. Overall, the readiness assessment underscored a strategic, phased approach to digital transformation, with leadership, collaboration, and workforce readiness forming the foundation for

Table 2. Digital transformation maturity and readiness assessment matrix.

Dimension	Operational definition	Readiness level (0-10)	Strategic importance (0-10)	Targeted minimum readiness	Readiness development priority	Resource allocation intensity level (0-10)
Collaboration Culture	The extent to which InnMedTech fosters cross-functional teamwork, knowledge sharing, and open communication among departments and external partners to facilitate digital transformation initiatives.	3.62	6	5	1	4.08
Data-Driven Services	InnMedTech's capability to leverage real-time data and advanced analytics to offer value-added healthcare services, such as remote patient monitoring and predictive maintenance for medical devices.	2.64	9	6	5	5.69
Employees Competence	The readiness of InnMedTech's workforce to adopt and effectively operate digital technologies through relevant skills, technical training, and continuous learning.	4.18	8	5	2	4.75
Leadership Support	The commitment of InnMedTech's senior management to drive digital transformation by providing clear strategic direction, allocating resources, and ensuring organisational alignment.	4.56	7	5	1	4.47
Smart Factory Technologies	The implementation of advanced manufacturing solutions, such as IoT, data analytics, and RPA, to create intelligent, adaptive production systems for healthcare product manufacturing.	1.88	10	5	3	6.44
Smart Operations	The integration of digital technologies into InnMedTech's production workflows to enable real-time monitoring, adaptive production planning, and data-driven decision-making.	3.38	7	6	4	5.63
Smart Products	The development of InnMedTech's healthcare devices with embedded sensors and connectivity, enabling remote monitoring, performance tracking, and user-driven customisation.	4.07	6	6	5	5.21
Strategic Alignment	The degree to which InnMedTech's digital transformation initiatives align with its overall business strategy, ensuring that technological advancements support long-term goals, sustainability, and regulatory compliance.	5.12	8	7	2	8.32
Value Network Integration	InnMedTech's ability to digitally connect with supply chain partners, healthcare providers, support service providers, and customers for real-time data exchange, collaborative production planning, and synchronised operations across the value chain.	2.94	9	5	4	7.67

advancing technology adoption, operational excellence, and value chain integration.

Stage two: Stakeholder collaboration and involvement

In *stage 2*, aligned with the ASM framework, InnMedTech prioritised stakeholder collaboration to ensure its digital transformation was guided by continuous engagement rather than technology alone. The company identified four key stakeholder groups: internal users, including leadership, production managers, and employees; external industry partners, such as healthcare providers, suppliers, and service providers; regulatory bodies to ensure compliance with healthcare and environmental standards; and end-users, including patients and consumer advocacy groups. A Stakeholder Influence-Interest Matrix was used to categorise stakeholders by their interest and influence, enabling targeted engagement strategies.

To operationalise stakeholder involvement, InnMedTech established a *stakeholder collaboration board*, comprising representatives from key stakeholder groups. Engagement was structured across three tiers: consultative, involving regular meetings, workshops, and forums to gather insights; participatory, through co-design sessions where stakeholders contributed to solution development; and collaborative, where stakeholders were actively involved in pilot projects and decision-making processes. This approach ensured that stakeholders were active contributors to the transformation journey.

To support effective collaboration, InnMedTech implemented a governance structure. A steering committee, led by company leadership, provided strategic oversight, while dedicated working groups focused on specific areas such as smart factory implementation, workforce upskilling, and sustainability practices. A digital collaboration platform facilitated transparent communication, document sharing, and progress tracking, ensuring stakeholders remained informed and engaged.

Clear, consistent communication was central to fostering trust and involvement. Internally, regular updates through emails and project meetings kept employees informed about progress and upcoming initiatives. Externally, InnMedTech maintained engagement through partner briefings and collaborative workshops with key stakeholders. Basic feedback mechanisms, such as periodic surveys and open forums, captured concerns and suggestions, ensuring stakeholder input guided project direction without overburdening resources.

To promote inclusive decision-making, InnMedTech adopted a collaborative decision-making framework (CDMF), ensuring that decisions were transparent, consensus-based, and evaluated against economic, environmental, and social outcomes. Stakeholders were engaged throughout the decision-making process, with outcomes documented and communicated via the common digital collaboration platform.

Stage three: Sustainability performance management system (SPMS)

To develop and implement the SPMS principle (capability) under the ASM framework, the company initiated a comprehensive sustainability performance audit to evaluate current operations and identify areas for improvement. This audit encompassed key performance indicators across energy consumption, material utilisation, workforce well-being, customer satisfaction, and compliance with environmental regulations. Each identified area was systematically mapped to potential technological interventions, ensuring that digital transformation efforts were strategically aligned with sustainability goals. This approach prioritised tangible, measurable benefits while proactively identifying and mitigating potential risks associated with new technologies.

To support technology adoption, InnMedTech applied a structured SWOT analysis for each proposed technology, leveraging the ASM knowledge repository to assess potential sustainability gains and avoid adverse effects. This repository provided insights into how various technologies impacted environmental, social, and economic performance, allowing the company to make informed decisions. For instance, while transitioning to automated production lines promised higher efficiency and reduced material waste, the analysis highlighted potential risks related to increased energy consumption if systems were not properly optimised. This informed the company's decision to implement energy-efficient automation solutions and integrate real-time monitoring systems to mitigate potential drawbacks.

In line with ASM's realism value, InnMedTech defined key performance indicators (KPIs) that were both achievable and realistic, ensuring that sustainability targets were grounded in operational feasibility. Rather than setting overly ambitious goals that could strain resources, the company focused on incremental improvements that could be sustained over time. KPIs included metrics such as energy efficiency per production unit, waste reduction percentages, workforce satisfaction scores, and adherence to ethical standards. A critical component of InnMedTech's approach was prioritising economic viability as a foundation for sustainability. Following the ASM philosophy, the company focused first on achieving economic gains through efficiency improvements, cost savings, and productivity enhancements without exploitation. These economic benefits were concurrently reinvested to advance social and environmental objectives. For example, savings from optimised production processes were allocated to workforce development

programmes and eco-friendly material sourcing initiatives, ensuring that sustainability efforts did not compromise financial stability.

To maintain accountability and drive continuous improvement, InnMedTech established a structured performance review cycle. Quarterly sustainability audits were conducted, where leadership and key stakeholders evaluated progress against defined KPIs. Insights from these reviews informed decision-making, allowing the company to adjust strategies, allocate resources effectively, and address emerging challenges. In parallel, the company promoted sustainability awareness across its workforce through targeted training programmes and internal communications. The aim was to ensured that employees at all levels understood the role of sustainability within the digital transformation journey and were equipped to contribute to continuous improvement efforts.

Stage four: New technology implementation projects

InnMedTech advanced to the fourth stage of digital transformation under the ASM framework after achieving maturity across key organisational areas. These included strengthened managerial competencies, targeted workforce skill development, and the readiness of foundational information and operational technologies (IT/OT). The foundation consisted of robust enterprise systems such as MES, ERP platforms, and communication networks. As the company progressed, it also invested in user interfaces, workflow automation, and advanced 3D printing systems to prepare both its front-end and back-end operations for the next phase of transformation.

The first major initiative was the deployment of IoT-enabled real-time monitoring across essential equipment and production processes. This project aimed not only to improve operational efficiency but also to support product innovation. InnMedTech developed smart, sensor-integrated medical devices that turned traditional products into intelligent, connected solutions. These devices enabled remote monitoring and predictive maintenance, which supported the company's shift toward a servitization model by adding service value to physical products and strengthening customer engagement.

The IoT implementation architecture was carefully planned with multiple integration layers to ensure smooth adoption. The design began with selecting high-precision, durable sensors and edge devices suited for industrial settings to support real-time data acquisition. Next, the composition of industrial control systems was optimised to manage equipment-level automation in line with the overall goals of the IoT project. Communication technologies were also specified. For instance, Ethernet was used in high-bandwidth, low-latency environments, while LPWAN was selected for remote asset monitoring. Communication protocols were standardised to align with system requirements, with AMQP used to support reliable message queuing. This comprehensive setup ensured cohesive functioning across all technical layers to enable continuous data collection, processing, and transmission.

An edge computing framework was also developed with well-defined technical specifications. The edge devices were designed to provide sufficient processing power, storage, and energy efficiency. For cloud infrastructure, InnMedTech adopted a hybrid cloud model. Private cloud services handled sensitive healthcare data, while public cloud platforms provided scalable storage and analytics capabilities. Middleware platforms were used to abstract and standardise data, enabling smooth integration between IoT devices and enterprise systems. This integration extended to the company's MES and ERP platforms, which acted as central hubs for data analytics, reporting, and process control. The overall architecture delivered end-to-end visibility and supported informed decision-making and continuous improvement.

In parallel with the IoT initiative, InnMedTech deployed robotics and automation technologies to manage intralogistics, part delivery, and material handling. The company integrated Automated Guided Vehicles (AGVs) to transport materials and implemented an Automated Storage and Retrieval System (ASRS) to improve inventory management. These systems were integrated with the warehouse management system (WMS) and ERP platform to coordinate operations across production and supply chains.

To increase administrative efficiency, InnMedTech introduced Robotic Process Automation (RPA) for back-end processes and repetitive managerial tasks. RPA was applied to automate invoice processing, shipment tracking, procurement approvals, inventory reconciliation, and HR-related documentation.

Operational intelligence was further enhanced through AI-powered analytics aimed at improving product management and supporting better decisions in sourcing, pricing, and material selection. Traditional methods based on historical data and manual analysis were inadequate for managing the complexity of healthcare manufacturing. To overcome this, InnMedTech established an AI analytics architecture that integrated real-time data pipelines from enterprise systems, IoT platforms, and external market data. In sourcing, AI algorithms evaluated supplier performance based on reliability, quality, and pricing history. This allowed the company to rank suppliers, negotiate better terms, diversify sources, and forecast material availability and cost

fluctuations. For pricing, AI-driven dynamic models continuously assessed production costs, market trends, and customer demand, enabling the company to adjust pricing without compromising profitability or satisfaction. In material selection, AI models were used to match product design requirements with available materials, identifying options that optimised quality, cost-efficiency, and sustainability.

All AI-driven insights were integrated into MES and ERP platforms through user-friendly dashboards that provided real-time metrics, forecasts, and actionable recommendations. The hybrid cloud setup supported scalable storage and high-performance computing. InnMedTech also emphasised transparency and explainability to ensure stakeholders understood the reasoning behind AI-generated suggestions.

Since simultaneous implementation of all these technologies would have been impractical and overly demanding, the company created separate technology roadmaps for each project. These roadmaps defined clear timelines, milestones, budget allocations, and technical guidelines to support phased deployment without disrupting current operations. The roadmaps also addressed the interdependencies among technologies. Each project was backed by detailed architectural designs covering infrastructure needs, data governance, and cybersecurity considerations to ensure successful and coordinated adoption.

Stage 5: Technology governance and sustainability orientation

Under the Technology Governance principle of ASM, InnMedTech established a Digital Transformation Office (DTO) to oversee all technology initiatives, ensuring that new systems were implemented with clear accountability, risk management, and alignment with business and sustainability objectives. A technology steering committee, composed of cross-functional leaders from IT, operations, procurement, and product development, evaluated each project based on technical feasibility, business impact, and potential sustainability contributions. This committee implemented a stage-gate evaluation process, ensuring that only technologies with clear value propositions advanced from pilot to full deployment.

To further streamline governance, the company implemented technology lifecycle management, covering evaluation, procurement, implementation, and end-of-life strategies. Each technology underwent SWOT analyses, not just for technical fit but also for sustainability trade-offs. This included evaluating whether energy-intensive solutions would undermine environmental goals, how workforce dynamics might shift, and whether supplier ecosystems were prepared for new standards. Data governance frameworks were also strengthened, particularly regarding IoT-enabled smart products that collected patient-related health data, ensuring GDPR compliance and ownership clarity while working with external platform providers.

Sustainability Orientation was integrated into both technology adoption and business model innovation, ensuring that sustainability was not treated as a parallel initiative but as an embedded outcome of digital transformation. To operationalise this principle, InnMedTech established a SPMS that tracked the environmental, social, and economic impacts of all digital initiatives. Unlike traditional sustainability reporting, the SPMS was designed to influence decision-making at both project and portfolio levels. For instance, when automating material handling with AGVs and ASRS, the assessment extended beyond operational efficiency to include lifecycle emissions, energy consumption, and material waste generated by the systems themselves.

To ensure sustainability gains were reflected in business value creation, InnMedTech conducted business model assessments alongside technology evaluations. This process explored how sustainability improvements, enabled by digitalisation, could become part of the company's value proposition. For example, IoT-enabled healthcare products were rebranded not only as advanced medical solutions but also as sustainability-driven offerings, emphasising energy-efficient production, eco-friendly materials, and extended product life cycles. This approach allowed the company to differentiate itself in the market. A crucial element of the Sustainability Orientation principle was the strategic reinvestment of economic gains from digital efficiencies into sustainability initiatives. InnMedTech allocated some of annual cost savings from operational improvements toward green projects, such as solar panel installations and eco-friendly packaging. Importantly, business model integration ensured that sustainability was not seen as a cost centre but as an opportunity for innovation and value creation.

Step Six: Advancing real-time and integration capabilities

Given the complexity and resource demands of real-time integration, InnMedTech prioritised critical areas where these capabilities were most impactful: production monitoring, equipment performance tracking, supply chain coordination, and product lifecycle management.

To enable real-time monitoring, the company deployed smart sensors across critical production assets, equipped with edge computing to process data locally before transmitting relevant insights to its cloud-based analytics platform. The purpose design of upgraded

industrial control systems facilitated real-time data acquisition, ensuring equipment-level data collection automation aligned across critical production workflows. The development of communication infrastructure was aligned satisfying with hardware requirement of real-time capability. For example, Ethernet-based Time-Sensitive Networking was used for high-priority data, Wi-Fi IEEE 802.11ax was used for flexible high-speed factory connectivity, while LPWAN was used for remote (often periodic) monitoring of less critical asset. Furthermore, real-time insights were centralised through a unified operations dashboard, integrated with InnMedTech's MES and ERP platforms. This dashboard provided live, digital twin-powered visibility into production progress, equipment status, and resource utilisation, allowing managers to identify bottlenecks or risk points and take immediate action. Real-time alerts ensured quick responses to anomalies, while historical trend analysis supported continuous improvement.

Achieving real-time insights also required seamless system interoperability across IoT platforms, MES, ERP, and third-party systems. To enable this, InnMedTech adopted a middleware-based integration architecture, connecting all platforms through APIs and Message Brokers like Apache Kafka and RabbitMQ, facilitating uninterrupted data flow. A standardised data exchange framework was implemented to ensure compatibility between machines, control systems, and enterprise platforms. This integration was further enhanced by Digital Twin visualisation technology, allowing virtual simulations of production workflows to test process changes and predict outcomes without disrupting operations.

External integration was equally prioritised, particularly for supply chain coordination and product lifecycle management. InnMedTech collaborated with supply chain partners to establish B2B integration platforms, enabling real-time data exchange for order management, inventory updates, and supplier collaboration. API-based integration facilitated real-time communication with suppliers and logistics providers, ensuring synchronised production planning, accurate inventory management, and timely deliveries. The success of this transformation relied on collaboration with technology vendors, system integrators, and consultants. Vendors supplied IoT devices, industrial control systems, and middleware platforms, while consultants ensured system compatibility, readiness, and alignment with ASM sustainability principles. System integrators played a pivotal role in developing the middleware framework, ensuring seamless communication between operational and enterprise platforms.

6.1. Project performance and outcome

InnMedTech's digital transformation, aligned with the ASM framework, led to measurable improvements across economic, social, and environmental dimensions. While the programme played a central role in these gains, other operational initiatives and external factors also contributed. Some initiatives initially caused short-term setbacks before delivering long-term value.

The most immediate improvements were seen in efficiency, resource use, and cost control. IoT-enabled monitoring and AI-supported decision-making raised OEE by 9.6 percent, while predictive maintenance cut unplanned downtime by 15 percent. Automation in material handling boosted warehouse throughput by 19 percent, and RPA reduced administrative processing times by 32.8 percent.

These gains translated into 5–7 percent annual cost savings and a 10 percent reduction in production cycle time. AI-supported pricing strategies added an estimated 2–3 percent to profit margins. However, during early implementation, production efficiency dropped by 5–7 percent as workflows adjusted and integration challenges disrupted operations. Warehouse automation also caused early inventory mismanagement and delays until system parameters were calibrated. By the end of 2024, inventory turnover had improved by 8.67 percent, and material waste was reduced by 11.07 percent due to automation and real-time tracking. The On-Time Delivery Rate rose to 88.13 percent, an 11 percent improvement since the programme began. Sales volume grew by 16.87 percent, and annual customer acquisition increased by 4.52 percent.

InnMedTech reinvested part of the economic gains into sustainability and workforce development, accepting short-term trade-offs for longer-term benefits. RPA and workflow optimisation reduced employee stress, while upskilling initiatives trained 21 percent of the workforce in digital tools and smart manufacturing. Although adoption varied across teams, employee satisfaction rose by 8 percent, reflecting increased engagement and digital readiness. Still, workforce adaptation challenges emerged during the transition.

Environmental outcomes also improved. IoT-based energy monitoring lowered energy use per production unit by 6–10 percent depending on product type. Material waste dropped by 6–8 percent, and logistics optimisation reduced transport-related emissions by 8.5 percent. Smart lighting and climate control systems further cut energy use. However, data inconsistencies across older production lines limited visibility and delayed progress toward energy and waste targets. For instance, while the company aimed to cut energy use by 15 percent, savings

Table 3. InnMedTech's IT performance indicators during the digital transformation project.

KPI	Description	Annual performance				Target performance
		2021	2022	2023	2024	
System Uptime (%)	Availability of critical IT/digital systems.	97.62	94.05	95.87	98.39	≥ 99.9%
IT Response Time (ms)	Average system response time.	536	419	218	196	≤ 200 ms
Digital Tool Adoption Rate (%)	Active usage rate of digital tools by employees.	76	74	81	88	≥ 85%
Digital Process Automation (%)	Share of processes automated digitally.	29	41	63	68	≥ 60–70%
User Satisfaction Score	Digital tool utility/usability rating.	82	69	78	90	≥ 85%
Real-Time Data Availability (%)	Critical data available in real-time for decisions.	49	68	82	85	≥ 90%
Data Accuracy Rate (%)	Accuracy of system-collected data.	88	92	85	96	≥ 95%
Predictive Maintenance Rate (%)	Predictive alerts preventing downtime.	37	68	72	81	≥ 85%
Mean Time Between Failures	Average time between system failures.	33.1	27.65	24.49	29.10	≥ 30 days
Mean Time to Recovery	Average recovery time post-failure.	2.18	1.98	2.45	2.42	≤ 1–2 h
IT Training Participation (%)	Percentage of critical users trained in digital tools.	~ 39	~ 62	~ 72	~ 88	≥ 80%

plateaued at 8–10 percent due to weak IoT integration in legacy systems. Despite efficiency gains, total energy consumption rose slightly due to the rebound effect, as increased output drove higher overall resource use.

A core strength of InnMedTech's transformation was the balance between economic performance and sustainability. Economic gains supported social and environmental programmes, creating a self-reinforcing cycle. For example, cost savings from AI-enabled efficiency were used to fund employee training, safety improvements, and green supply chain efforts. Supplier engagement also improved. By the end of the project, 28 percent of Tier-1 suppliers were integrated into the company's real-time supply chain platform. However, many suppliers lacked the technical readiness for real-time data exchange, which caused occasional procurement delays and disruptions during the first two years.

From a technical standpoint, InnMedTech's IT performance showed steady improvement throughout the digitalisation process. As Table 3 shows, early challenges in reliability, user satisfaction, and training were gradually addressed as the company gained experience. Key indicators such as system responsiveness, tool adoption, process automation, and data quality improved steadily. Predictive maintenance and real-time analytics became more effective, boosting decision-making and operational foresight. Workforce development efforts paid off, enhancing digital system adoption across functions. Some areas, such as system uptime and recovery speed, still require optimisation to meet top-tier benchmarks. Overall, the company demonstrated a well-managed digital transition characterised by growing resilience, operational efficiency, and alignment with long-term strategic goals.

6.2. Benchmarking ASM performance against smart manufacturing

This section presents a data-driven comparison of ASM and Smart Manufacturing, using longitudinal performance data from the InnMedTech case study. The timeframe covers 2013–2024, starting from the point at which consistent data became available for the selected indicators. The timeline includes the pre-digital phase (2013), the Smart Manufacturing transition (2014–2017), Smart Manufacturing maturity (2018–2020), and the ASM transition phase (2021–2024) within the case company.

Sixteen key performance indicators (KPIs) are used to evaluate operational efficiency, reliability, responsiveness, and social sustainability. Table 4 presents the full performance matrix based on the InnMedTech case, showing year-by-year values for all KPIs across the different transformation phases. The values reflect both absolute performance levels and the broader evolution of the firm's digital manufacturing capability. The data shows that Smart Manufacturing delivered clear gains across many technical and operational indicators. For instance, labour productivity increased from 179.8 units per FTE in 2013–252.8 by 2020. Defects per unit fell from 5.6–3.11 during the same period, while time to market dropped from 23.3–16.6 weeks. MTBF (Mean Time Between Failures) improved from 410 to 837 h. These outcomes highlight the effectiveness of Smart Manufacturing in stabilising production, increasing automation, and reducing waste.

Starting in 2021, ASM was implemented as a continuation of this transformation, and the performance trendlines shifted. In most operational indicators, ASM either maintained or slightly improved performance. Labour productivity reached 277.4 units per FTE in 2024, and time to market was further reduced to 14.3 weeks. MTBF increased to 1029 h, and defects per unit dropped from 3.11 in 2020–2.61 in 2024. This confirms that ASM does not compromise the operational gains achieved through Smart Manufacturing. Instead, it reinforces and stabilises them.

Where ASM provides a clearer performance advantage is in social and workforce-related dimensions. Employee satisfaction rose from 76.3 in 2020–85.3 in 2024, and the employee turnover rate dropped from 16.5 percent to 8.1

Table 4. Yearly performance trends across Smart Manufacturing and ASM in case company.

Year	Stage	Defects per Unit	Customer Complaints Ratio (%)	Scrap Level (%)	Warranty Costs Ratio (%)	MTBF (hrs)	Time to Market (weeks)	Labour Productivity (units/FTE)	Capacity Utilisation Rate (%)	Employee Satisfaction Score (ESAT)	Employee Turnover Rate (%)	Productivity per Hour (units/hr)	Mean Time to Recovery (MTTR, hrs)	Production Downtime Ratio (%)	Gender Pay Gap (%)	Skill Pay Gap (%)
2013	Pre-Smart manufacturing	5.60	4.55	5.84	3.02	410	23.3	179.8	63.4	62.6	13.4	15.8	12.25	12.17	17.46	42.31
2014	Smart manufacturing Transition	5.11	3.82	5.65	2.60	420	22.4	199.6	66.1	64.8	16.9	15.5	11.16	9.93	16.93	37.79
2015	Smart manufacturing Transition	4.90	2.95	4.89	2.47	555	19.8	190.5	71.0	66.6	13.1	19.1	10.78	9.34	14.49	37.46
2016	Smart manufacturing Transition	4.71	3.49	4.97	2.42	607	18.4	219.6	71.9	68.3	15.0	17.7	10.44	9.42	14.83	38.48
2017	Smart manufacturing Transition	4.00	3.1	5.02	2.27	601	20.1	227.7	72.3	68.3	14.1	21.0	8.79	8.38	12.67	38.02
2018	Smart manufacturing Maturity	3.63	3.26	4.86	2.02	782	19.7	237.6	76.5	72.5	16.7	24.0	10.02	8.58	13.8	41.89
2019	Smart manufacturing Maturity	3.63	2.52	3.53	1.79	867	16.3	251.4	77.7	75.0	15.7	22.2	12.15	8.02	14.31	43.19
2020	Smart manufacturing Maturity	3.11	2.42	3.88	1.57	837	16.6	252.8	81.6	76.3	16.5	24.0	11.97	7.92	14.58	41.64
2021	ASM Transition	3.25	2.39	2.94	1.25	857	14.4	243.4	77.3	79.9	13.9	23.2	9.58	8.21	11.53	39.60
2022	ASM Transition	3.24	1.75	2.25	1.13	945	15.6	258.1	78.2	83.3	11.2	24.8	8.32	8.03	10.31	32.47
2023	ASM Transition	2.87	1.51	2.43	0.95	937	14.8	265.6	82.1	84.5	9.70	26.1	8.04	7.77	9.70	28.58
2024	ASM Transition	2.61	1.19	2.41	0.91	1029	14.3	277.4	84.1	85.3	8.10	28.1	7.27	6.93	8.78	22.59

percent. The gender pay gap declined from 14.58 percent to 8.78 percent, and the skill pay gap narrowed from 41.64 percent to 22.59 percent over the same period. Production downtime also continued to decrease, from 7.92 percent in 2020–6.93 percent in 2024, and mean time to recovery improved from 11.97–7.27 h, indicated clear enhancement of the company's manufacturing resilience capability.

While not all improvements can be solely attributed to Smart Manufacturing or ASM, as external factors and accumulated experience contribute as well, the data supports two clear interpretations. First, ASM consolidates and sustains the efficiency gains of Smart Manufacturing. Second, it expands the transformation agenda to include equity, employee well-being, and human-system interaction as key performance areas. The absence of performance degradation across any major indicator further reinforces ASM's viability as a next-stage digital production framework aligned with Industry 5.0 priorities.

6.3. Challenges and pitfalls

Despite its structured and phased approach, the implementation of the ASM framework at InnMedTech faced several noteworthy challenges that reflect the broader complexities of transitioning toward Industry 5.0. One of the foremost challenges was the scalability of digitalisation efforts across different functional units. While pilot projects and isolated automation deployments demonstrated promising results, extending these technologies uniformly across the entire value chain proved difficult due to system compatibility issues, legacy infrastructures, and varying digital maturity levels among internal departments and external partners. This limited the seamless realisation of ASM's integration and real-time capabilities in the early phases.

Cost considerations presented another persistent obstacle. The investments required for advanced technology acquisition, upskilling programmes, and infrastructure modernisation were substantial. Although the company experienced notable economic gains, the initial capital expenditure and ongoing maintenance costs placed a significant burden on financial planning. These costs were especially challenging given the need to demonstrate short-term returns to stakeholders while pursuing long-term sustainability goals, underscoring a tension often encountered in transitioning to Industry 5.0.

Intra-organisational resistance further complicated implementation. Employees expressed concerns about job displacement, increased complexity, and technostress associated with new tools and workflows. Even with change management initiatives in place, resistance to adopting unfamiliar digital processes led

to delays in adoption and inconsistent performance improvements. Moreover, aligning diverse stakeholder interests within the organisation required significant cultural shifts toward collaboration and continuous learning.

Beyond these core challenges, strategic alignment and roadmap execution required ongoing adjustments. Despite the structured roadmap laid out during the DSMC phase, shifting priorities and emergent operational issues demanded frequent recalibration. Ensuring that the roadmap remained flexible yet strategically coherent tested the company's governance structures and planning competencies.

Another major challenge stemmed from data governance and interoperability. With the integration of IoT systems, predictive analytics, and cloud platforms, InnMedTech encountered difficulties in ensuring secure, accurate, and interoperable data flows across systems and partners. Data silos persisted in several parts of the organisation, limiting real-time visibility and undermining decision-making effectiveness. These technical challenges also posed risks to sustainability reporting accuracy and compliance obligations.

Finally, external stakeholder engagement, including collaboration with regulators and industry partners, revealed systemic limitations in the broader digitalisation ecosystem. The pace of regulatory adaptation lagged behind technological innovation, and misaligned digital standards across partners hindered supply chain integration. Despite InnMedTech's proactive engagement efforts, a lack of digital readiness among smaller supply chain partners introduced bottlenecks in implementing end-to-end digital solutions, highlighting the importance of industry-wide synchronisation for effective ASM adoption.

These challenges underscore that while ASM provides a valuable framework for balancing technology and sustainability, its successful application depends on overcoming significant structural, cultural, and systemic barriers. Addressing these pitfalls requires not only organisational commitment but also coordinated action across value networks and policy environments.

7. Discussion

The InnMedTech case shows how ASM enabled a balanced digital transformation by aligning technological progress with business goals, workforce development, and environmental responsibility. ASM aims to prevent the consequences of poorly governed technological integration, which can negatively affect economic and socio-environmental performance. The InnMedTech case demonstrates how ASM's structured governance and sustainability focus helped monitor and adjust transformation efforts. The framework emphasises that technologies like AI and C-CPS must be managed proactively to avoid unintended harm.

ASM can be disruptive, requiring significant changes across manufacturing value chains. The interplay of enablers and preventers is central to this transformation, with core technologies acting as moderators. The case study showed how InnMedTech followed a staged approach, first developing digitalisation strategic management capability and digitalisation resource readiness. These capabilities enabled key preventers, including stakeholder collaboration and a sustainability performance management system. These preventers ensured that digital initiatives stayed aligned with sustainability goals and maintained transparency.

Building on this foundation, InnMedTech advanced two dependent preventers: technology governance and sustainability orientation. These guided the ethical and societal management of new technologies while supporting economic outcomes. The firm used AI and IoT not only for efficiency but also for sustainability, applying real-time energy monitoring and predictive maintenance to reduce waste. This illustrates how technology governance, when applied practically, helps minimise risks and maximise positive outcomes.

Once these capabilities were in place and core technologies integrated, InnMedTech developed real-time and integration capabilities. Real-time capability enabled responsive monitoring and mitigation of inefficiencies and sustainability risks. Integration capability ensured seamless data flow across operations and supply chains, improving collaboration and transparency. While the company initially faced challenges like fragmented workflows and inconsistent data, increased digital maturity improved system interoperability and sustainability performance.

The case confirms that sustainable digital transformation is not a linear process. It requires stepwise progression and continuous adaptation. The mutual dependence of enablers and preventers means progress in one area often depends on another. At InnMedTech, resource readiness enabled stakeholder collaboration, which in turn supported sustainability management and technology governance. This foundation allowed real-time and integration capabilities to mature, illustrating the iterative nature of ASM.

InnMedTech's ongoing programme reflects ASM's long-term view. Transformation is not a one-time shift but an evolving process. The company applied feedback loops, audits, and stakeholder consultations to ensure ongoing alignment with business and sustainability goals. This continuous improvement cycle illustrates how ASM

supports resilience in an evolving technological and regulatory landscape. Ultimately, ASM, when used as a staged digital transformation farmwork, can equip manufacturers to manage the complexities of digital transformation while advancing inclusive sustainability. InnMedTech's ability to somewhat balance profitability, environmental goals, and employee well-being highlights the framework's value as a practical governance model.

While grounded in responsible governance and sustainability, ASM also recognises the ethical and social trade-offs of digitalisation. It aims to align technological advancement with human-centric values, balancing performance, environmental care, and workforce needs. Yet, risks remain. Technologies like AI and robotics may introduce challenges such as bias, surveillance, or opaque decision-making. Automation may displace jobs, particularly in low-skill roles. Although ASM promotes upskilling and learning, success depends on a company's ability to invest in human capital and lead organisational change. ASM treats these risks as design factors, emphasising governance and training to limit negative impacts. Still, challenges like job loss, ethical AI use, and data privacy cannot be fully eliminated, especially in complex industrial contexts. Therefore, ASM presents transformation as a continuous, adaptive process that must remain responsive to emerging ethical and social concerns.

8. Concluding remarks

The present study drew on Industry 5.0 scientific background and introduced ASM as an approach for leveraging digital technologies to pursue inclusive sustainable manufacturing. The preventers and enablers discussed in this study serve as crucial mechanisms to guide organisations on this transformative journey, allowing them to unlock the potential benefits of these technologies while minimising any adverse sustainability impacts. However, it is essential to recognise that the challenges of harmonising economic growth, social well-being, and environmental preservation within the dynamic landscape of Industry 5.0 require ongoing attention and adaptation. The success of ASM in creating an idealistically sustainable industry hinges on addressing these challenges effectively. This study marks an early step in that direction, with much more to be explored and achieved in the ASM context. The present work is believed to offer notable implications for theory, practice, and future research.

8.1. Contribution to research

The ASM framework contributes to the ongoing scholarly discourse on sustainable digital transformation by providing a structured, adaptive framework that bridges the gap between technological advancement and sustainability goals in the context of Industry 5.0. At its core, ASM challenges the traditional dichotomy between productivity-driven digitalisation under Industry 4.0 and the idealistic sustainability aspirations of Industry 5.0, presenting a pragmatic middle ground where technology adoption is governed by sustainability principles. The case study of InnMedTech demonstrates how ASM operationalises this balance, offering empirical evidence that reinforces its theoretical foundations.

First, ASM advances the capability-based perspective in digital transformation literature by emphasising the interdependencies between core technologies, enablers, and preventers. Unlike existing frameworks that treat digitalisation as a linear, technology-driven process, ASM recognises the hierarchical nature of capability development. It posits that manufacturers must first build foundational competencies, such as DSMC and DRR, to enable the development of more complex capabilities like Stakeholder Collaboration, Sustainability Performance Management, Technology Governance, and Sustainability Orientation. This sequential yet iterative approach expands the dynamic capabilities theory by demonstrating how capability accumulation and feedback loops drive continuous transformation rather than static maturity.

Second, ASM introduces a socio-technical systems perspective, emphasising the mutual shaping of technology, organisational practices, and social structures. This addresses a critical gap in Industry 4.0 literature, which often overlooks the social implications of digital transformation. The case study at InnMedTech highlighted how the ASM framework facilitated cross-functional collaboration, upskilling initiatives, and stakeholder engagement, ensuring that technological advancements were aligned with workforce well-being and social inclusion. This human-centric approach aligns with Industry 5.0 ideals, providing a conceptual foundation for scholars exploring responsible digitalisation and ethical technology adoption in manufacturing.

Third, ASM enriches sustainability-oriented innovation literature by presenting a practical pathway for manufacturers to achieve socio-environmental sustainability through technology-driven transformation. While existing frameworks like Industry 5.0 emphasise human-centricity and resilience, they often lack operational mechanisms for implementation. ASM addresses this gap by operationalising sustainability governance through principles such as Technology Governance, Sustainability Performance Management, and Real-Time Monitoring, ensuring that technology adoption is continuously evaluated against environmental and social

performance metrics. The InnMedTech case evidenced how this approach enabled the company to balance economic gains with environmental stewardship and workforce well-being, offering empirical support for ASM's theoretical propositions.

Fourth, ASM contributes to the governance of digital ecosystems, highlighting the role of collaboration among manufacturers, technology providers, regulators, and other stakeholders. This aligns with emerging ecosystem-based frameworks in digital transformation literature but extends their scope by emphasising the sustainability implications of ecosystem governance. The InnMedTech case illustrated how cross-functional decision-making, stakeholder engagement, and transparent data governance under ASM strengthened trust and accountability, reinforcing the importance of multi-stakeholder governance in sustainable digital transformation.

8.2. Contribution to practice

The ASM framework provides pragmatic, actionable guidance for manufacturers navigating the complexities of digital transformation while pursuing sustainability goals. The case study of InnMedTech illustrates how ASM principles can be practically implemented, offering valuable lessons for practitioners seeking to balance technological advancement with socio-environmental responsibility.

First, ASM offers a structured, stage-gated approach to digital transformation, emphasising the sequencing of capabilities based on organisational readiness and strategic priorities. Manufacturers are guided to begin with foundational capabilities, such as Digitalization Strategic Management and Resource Readiness, before progressing to more complex initiatives. The InnMedTech case demonstrated how this phased approach prevented overextension, ensuring that digital initiatives were implemented realistically, with achievable goals and measurable outcomes. This approach reduces the risk of technology failures, budget overruns, and disrupted workflows.

Second, ASM promotes proactive technology governance, ensuring that digital transformation efforts are ethically sound, socio-environmentally responsible, and economically viable. The InnMedTech case showed how technology steering committees, regular performance audits, and risk assessments enabled responsible technology adoption, mitigating negative externalities while maximising sustainability gains. This approach empowers practitioners to adopt emerging technologies like AI, IoT, and automation without compromising ethical standards or regulatory compliance.

Third, ASM provides actionable strategies for achieving sustainability-oriented innovation, ensuring that digital transformation efforts contribute to environmental stewardship, social well-being, and economic resilience. At InnMedTech, the SPMS enabled the company to track sustainability metrics, ensuring that energy efficiency, waste reduction, and workforce well-being were integrated into technology adoption decisions. This approach ensured that sustainability was not treated as a standalone objective but was embedded into core business processes, providing practitioners with clear, measurable goals for sustainable digitalisation.

Fourth, ASM highlights the importance of real-time capability and integration capability as critical enablers for agile decision-making, operational resilience, and supply chain optimisation. At InnMedTech, the adoption of IoT-enabled monitoring systems, automated workflows, and AI-powered analytics facilitated data-driven decision-making, improving operational efficiency while enhancing sustainability outcomes. This integrated approach enabled the company to respond quickly to emerging challenges, such as supply chain disruptions and energy management issues, demonstrating the practical value of real-time insights.

Fifth, ASM promotes cross-functional collaboration and stakeholder involvement, ensuring that digital transformation efforts are inclusive, transparent, and socially responsible. The case study illustrated how multi-stakeholder engagement, cross-functional teams, and transparent communication channels fostered shared ownership of digital initiatives, enhancing employee engagement, regulatory alignment, and customer satisfaction. This approach helps practitioners navigate the complexities of stakeholder management, ensuring that digital transformation efforts align with broader societal expectations.

Sixth, ASM emphasises the importance of workforce skill development, ensuring that employees remain adaptable and resilient in the face of technological change. At InnMedTech, targeted upskilling programmes, digital literacy initiatives, and collaborative learning environments empowered employees to engage effectively with new technologies, reducing resistance to change while enhancing job satisfaction and workforce retention. This approach offers practitioners practical guidance for addressing skill gaps, ensuring that human capital remains a core enabler of sustainable digital transformation.

Ultimately, ASM provides practitioners with a structured, adaptive framework for achieving sustainable digital transformation, ensuring that technological advancement, sustainability governance, and stakeholder engagement are mutually reinforcing. By demonstrating how

ASM can be operationalised within a real-world manufacturing context, the InnMedTech case offers practical insights for other manufacturers seeking to navigate the Industry 5.0 landscape while advancing sustainable, socially responsible manufacturing practices. Furthermore, the ASM framework and the case study provide valuable practical insights, summarised as follows:

- Digitalisation should proceed incrementally, targeting specific technologies and processes while ensuring the organisation possesses adequate capability readiness. Leapfrogging without sufficient preparation often results in failure.
- Resistance to change and skill gaps among employees may persist despite strong communication and training efforts. The focus should remain on realistic mitigation of job displacement, skill mismatches, and technology-induced stress.
- Sustainability goals should be defined pragmatically, grounded in operational feasibility, and supported by economic gains generated through digitalisation initiatives.
- Digital transformation (under Industry 4.0/5.0 or similar paradigms) is most effective when approached as a continuous, evolving programme rather than a one-time project, with periodic reassessment and adaptation of strategies.
- Success depends on prioritising technology adoption based on organisational readiness, sustainability impact, and economic feasibility, rather than pursuing innovations solely for their novelty.
- Challenges and obstacles encountered during digital transformation should be viewed as manageable risks rather than indicators of failure, requiring strategic mitigation and adaptive steering.
- Effective digitalisation requires cross-functional teams, ensuring that technology, operations, sustainability, and workforce needs are addressed collectively.
- Clear governance structures and data ownership policies are crucial when collaborating with external technology providers, particularly in regulated industries.
- While digitalisation can enhance efficiency, it may also lead to increased resource consumption, requiring proactive sustainability strategies to mitigate rebound effect and unintended impacts.

8.3. Limitation and future direction

While ASM offers a structured and integrative approach to sustainable digital transformation, its validation in this study is based on a single in-depth case of a digitally maturing healthcare manufacturer. As such, the generalizability of the findings is inherently limited. The InnMedTech case reflects the realities of a mid-sized firm operating in a regulated, high-tech environment, which may not be representative of other sectors, firm sizes, or regions with different institutional and technological conditions. Broader validation of ASM requires comparative studies across diverse industrial contexts to understand how the framework adapts to variations in organisational scale, market dynamics, and regulatory landscapes.

Future research should focus on multi-case, cross-sectoral investigations to assess how ASM performs in different manufacturing domains, including resource-constrained small enterprises and less digitised sectors. Comparative benchmarking based on larger datasets would help identify recurring implementation patterns, critical success factors, and context-specific challenges, providing a stronger empirical basis for refining and tailoring the framework.

Furthermore, while this study incorporated operational KPIs to evaluate economic and environmental improvements, the empirical validation of ASM remains at an early stage. Future work could develop quantitative models for assessing ASM's impact using standardised sustainability performance indicators. Such tools would help organisations track progress, identify gaps, and benchmark efforts in a structured and consistent way.

Given the growing complexity of digital transformation, future research should examine the socio-technical and organisational barriers that may hinder ASM implementation. These may include workforce resistance, budgetary constraints, integration difficulties, and regulatory misalignments at both national and international levels. Longitudinal studies can shed light on how firms navigate these barriers over time, highlighting the enablers and mitigation strategies that underpin successful outcomes. Since ASM depends heavily on stakeholder collaboration, it is also important to explore socio-cultural dynamics such as trust-building, organisational culture, and cross-functional communication. These factors shape how technological initiatives are received and sustained, and they are critical for aligning digital transformation with long-term sustainability goals.

At the same time, ASM must remain responsive to evolving technological paradigms. As AI, robotics, and data-driven automation become more embedded in industrial settings, future research should assess how ASM can adapt without compromising its human-centric values. Ethical issues such as algorithmic bias, explainability, data privacy, and workforce displacement need to be addressed through governance mechanisms that reflect ASM's commitment to responsible innovation. Expanding the scope of ASM beyond the factory

floor is another key direction. Applying its principles across the broader value chain, including sourcing, logistics, and supplier engagement, requires interoperable digital infrastructures and governance structures that enable transparency and cross-organisational collaboration. Finally, future work should examine how ASM aligns with the broader ambitions of Industry 5.0. Studying the intersections between their values, technologies, and implementation models will help clarify how ASM can support balanced, inclusive, and ecologically sound industrial progress.

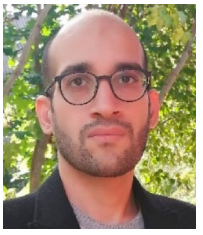
Declaration

While preparing this work, the authors used an AI-based editing tool (ChatGPT 4.0) to proofread some parts of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take responsibility for the publication's content.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors



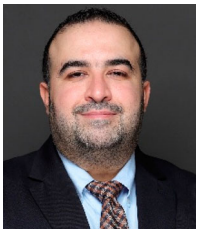
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Data availability statement

The data that support the findings of this study are available from the corresponding author, Dr. Morteza Ghobakhloo, upon reasonable request.

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